Semianalytic Reconstruction of the Mean-Field Gain Model and the Spectral Fluctuation Theory for ^{85}Rb Double-Λ Four-Wave Mixing

Atomic response, microscopic noise, transport, and detected spectra

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Scope and status. This study separates semiclassical gain, microscopic quantum fluctuations, transported atomic ensembles, and external detection. It derives the atomic GKSL drift and ordered diffusion, their independent quantum regression reference, reciprocal field normalization, physical sideband channels, periodic and spatial response, finite-age boundary noise, thermal inflow and nonlocal propagation requirements. These constructions yield conditional spectra for declared reduced models. They do not yet establish a converged thermal-vapor cell spectrum or quantitative apparatus agreement.
The phenomenological gain model and reciprocal microscopic model are distinct model variants. The former gives $G_p = 403.53550$ at one measured operating point and substantially disagrees with the measured output powers. A gain-only score or an independent-vacuum completion is not a microscopic squeezing prediction. The thermal inflow calculation passes selected path accuracy gates but all six directed comparisons between three coarse ensemble samples fail a 5% gate; source-resolved differences reach 43.61%. Boundary preparation, ensemble refinement, nonlocal Maxwell closure, full Zeeman physics and measured detector inputs remain material conditions on any apparatus claim. All numerical values below state the model and test to which they belong.

Abstract

We derive a theory of seeded warm-vapor ^{85}Rb D1 double-Λ four-wave mixing from the density-matrix generator to SQL-normalized detection. The stationary response is a projected Liouvillian resolvent, with a simple-pole expansion for diagonalizable generators and a Jordan/contour qualification at defective points. The distributed field-noise integral is an entire function of a Lyapunov superoperator, evaluated without singular eigenvector coordinates. Explicit conventions separate angular and ordinary frequencies, optical and analysis detunings, power and photon-flux gain, and atomic from spatial field diffusion.

Microscopic ordered diffusion follows from the adjoint GKSL product rule. Its resolvent spectrum is independently checked by full-density-matrix quantum regression. Reciprocal dipole normalization exposes the limitations of a common scalar hyperfine factor in a pumped four-state surrogate. Physical positive/negative sidebands, Gaussian complete positivity, finite RF filters, periodic seeds, two moving optical phases and finite spatial collection modes are treated explicitly. Ballistic atoms carry initial quantum noise and jump noise across positions; Poisson entry adds number noise. An adjoint trajectory construction supplies a source-resolved reference, while exact constant-step convolutions and a fourth-order composition control numerical error along each path.

Numerical examples distinguish admissibility, discretization convergence and apparatus validation. The atomic Raman half-width 0.520 MHz at one zero- velocity operating point and the angular Doppler rms width 1.38 MHz do not determine a squeezing bandwidth. A positive finite Floquet state can still have a 16.8% gain truncation error. Removing transported boundary noise can retain PSD noise while changing a spectrum by 14.59%. Selected trajectory solutions

converge, but three independent coarse thermal samples disagree by as much as 43.61% in source-resolved covariance. The resulting theory supports conditional reduced-model calculations and explicit counterexamples; a converged, measured-input cell squeezing prediction remains unestablished.

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Part I
Definitions and conventions

1 Atomic references and frequency definitions

The four hyperfine states are

$$g_1 = 5S_{1/2}(F=2), \quad g_2 = 5S_{1/2}(F=3), \quad e_2 = 5P_{1/2}(F'=2), \quad e_3 = 5P_{1/2}(F'=3).$$

The optical reference is

$$\omega_{13} \equiv (E_{e_3} - E_{g_1})/\hbar, \qquad \omega_{\mathrm{HF}} \equiv (E_{g_2} - E_{g_1})/\hbar.$$

Every detuning and linewidth in an equation is an *angular* frequency. Numerical values are always displayed after division by 2π:

$$\frac{\Delta}{2\pi} = -1.50\,\mathrm{GHz}, \qquad \frac{\delta}{2\pi} = -280\,\mathrm{MHz}, \qquad \frac{\Gamma}{2\pi} = 5.746\,\mathrm{MHz}.$$

The pump and probe-seed conventions are

$$\Delta = \omega_{\mathrm{pump}} - \omega_{13}, \tag{1}$$
$$\omega_p = \omega_{\mathrm{pump}} - \omega_{\mathrm{HF}} + \delta, \tag{2}$$
$$\Omega_{\mathrm{beat}} = \omega_{\mathrm{pump}} - \omega_p = \omega_{\mathrm{HF}} - \delta, \tag{3}$$
$$\omega_c = 2\omega_{\mathrm{pump}} - \omega_p. \tag{4}$$

Thus Ω_{beat} is the optical pump–probe-seed beat that organizes the Floquet harmonics. It is not the analysis frequency.

The independent radio-frequency analysis variable is

$$\frac{\Omega_{\mathrm{SA}}}{2\pi} \simeq 0.1\text{–}4\,\mathrm{MHz}.$$

It labels quantum sidebands at $\omega_p \pm \Omega_{\mathrm{SA}}$ and $\omega_c \mp \Omega_{\mathrm{SA}}$. With the Fourier convention used here,

$$\delta\hat{a}_j(z,t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega_{\mathrm{SA}}}{2\pi}\, \delta\hat{a}_j(z,\Omega_{\mathrm{SA}}) \mathrm{e}^{-\mathrm{i}\Omega_{\mathrm{SA}} t}. \tag{5}$$

Table 1: Frequency definitions used throughout the report.

Symbol	Definition	Reported as	Role
ω_{13}	$(E_{e_3} - E_{g_1})/\hbar$	$\omega_{13}/2\pi$	atomic optical reference
Δ	$\omega_{\mathrm{pump}} - \omega_{13}$	$\Delta/2\pi$	pump one-photon detuning
δ	$\omega_p - \omega_{\mathrm{pump}} + \omega_{\mathrm{HF}}$	$\delta/2\pi$	Raman detuning
Ω_{beat}	$\omega_{\mathrm{pump}} - \omega_p$	$\Omega_{\mathrm{beat}}/2\pi$	optical pump–probe-seed beat
Ω_{SA}	sideband offset from an optical carrier	$\Omega_{\mathrm{SA}}/2\pi$	RF analysis (sideband) frequency
ω_p, ω_c	probe and conjugate carrier angular frequencies	$\omega_{p,c}/2\pi$	optical carriers with $\omega_c = 2\omega_{\mathrm{pump}} - \omega_p$
$-\operatorname{Im}\lambda_m$	isolated-pole center in the convention of Eq. (5)	$-\operatorname{Im}\lambda_m/2\pi$	compared with $\Omega_{\mathrm{SA}}/2\pi$, never with $\Omega_{\mathrm{beat}}/2\pi$
Γ	population-decay angular rate	$\Gamma/2\pi$	natural linewidth

Mode labels. The three optical fields are the *pump*, the *probe* (Stokes), and the *conjugate* (anti-Stokes); the weak field injected on the probe leg is the *probe-seed*, and probe and conjugate together are the *twin beams* [1]. Subscripts follow that naming: p denotes the probe and c the conjugate, while pump quantities carry the explicit subscript ″pump″ so that no symbol is overloaded. Powers are P_{pump}, P_0 for the injected probe-seed, and P_p, P_c for the two output beams. Macroscopic polarizations are written $\mathcal{P}_j$ to keep them distinct from powers, and the quadratures of mode j are (X_j, Y_j).

2 Field, beam, and susceptibility normalization

Canonical traveling-wave amplitudes are photon-flux normalized:

$$[\delta\hat{a}_j(z,t), \delta\hat{a}_k^\dagger(z,t')] = \delta_{jk}\delta(t-t'), \qquad P_j = \hbar\omega_j|\alpha_j|^2. \tag{6}$$

If the atomic code uses Rabi amplitudes $\Omega_j = \mathcal{G}_j\alpha_j$, its propagation matrix must be transformed before being interpreted quantum mechanically:

$$M_{\mathrm{can}} = \mathcal{G}^{-1}M_\Omega\mathcal{G}, \qquad \mathcal{G} = \mathrm{diag}(\mathcal{G}_p, \mathcal{G}_c^*). \tag{7}$$

The near equality of the two optical wavelengths does not remove this normalization requirement. Every quantity below that is contracted with the canonical metric $J = \mathrm{diag}(1,-1)$ — in particular the commutator-defect matrix of Sec. 16 — is basis dependent through Eq. (7) and must be recomputed if the collected mode areas differ.

The quoted beam sizes

$$w_{\mathrm{pump}} = 530\,\mu\mathrm{m}, \qquad w_p = 330\,\mu\mathrm{m}$$

are $1/\mathrm{e}^2$ *intensity radii*, not diameters.

The reduced response is defined schematically by $\bar{\chi}_{jk} = \rho_{eg}^{(j\leftarrow k)}/\Omega_k$. Consequently

$$[\bar{\chi}_{jk}] = \mathrm{s}, \qquad \chi_{jk} = -\frac{2Nd^2}{\epsilon_0\hbar}\,\ell_{\mathrm{norm}}\bar{\chi}_{jk}, \qquad [\chi_{jk}] = 1. \tag{8}$$

The factor ℓ_{norm} is dimensionless. Its decomposition is audited in Appendix E. The phenomenological convention uses $1/[2(2I+1)] = 1/12$ together with trace-one manifold populations. This removes one explicit population duplicate but is not the manifold-resolved weak-absorption normalization (Sec. 22); 0.74 remains an assumed residual.

3 Operating-point classification

Table 2: Operating points are classified by provenance, not by numerical appearance in a scan.

Point and class	$\Delta/2\pi$	$\delta/2\pi$	T	Meaning
Archived reduced-model optimum	−1.50 GHz	−280 MHz	110 °C	Numerical optimum of the gain-only squeezing objective. It is not an experimental optimum and its $N_F = 1$ result is unconverged.
Sim et al. literature benchmark [1]	+0.9 GHz	−8 MHz	121 °C	Experimentally measured operating point: 600 mW pump, 8–10 μW probe-seed, 12.5 mm cell, $w_{\mathrm{pump}} = 530\,\mu$m, $w_p = 330\,\mu$m, and $\theta = 0.32°$.

The literature describes the gain as about 15; the cited ^{85}Rb squeezing trace reports raw powers $(P_0, P_p, P_c) = (8, 111, 109)\,\mu$W, which give the power ratios 13.875 and 13.625 used here. It reports maximum IDS of −7.8 dB, an approximate 3.5 MHz bandwidth, and a laser-source electronic-noise peak near 2 MHz. Those measurements are not properties of the diagnostic $(\Delta/2\pi, \delta/2\pi) = (-1.50\,\mathrm{GHz}, -280\,\mathrm{MHz})$ model point.

Part II
Pump-only atomic steady state

4 Self-consistent finite-dimensional Liouvillian

For each velocity class and longitudinal position, the pump-only state is defined by

$$\mathcal{L}_{\rm pump}(z,\boldsymbol{v})\rho_{\rm ss}(z,\boldsymbol{v})=0, \qquad \mathrm{Tr}\,\rho_{\rm ss}(z,\boldsymbol{v})=1. \tag{9}$$

For the standard minus Raman branch, the seeded frame has $\Omega_{\rm beat}=\omega_{\rm HF}-\delta$. The explicit transformation

$$U_-(t)=\exp(+\mathrm{i}\Omega_{\rm beat}t|g_2\rangle\langle g_2|) \tag{10}$$

is used with $\rho_{\rm pump}=U_-^\dagger\rho_{\rm seed}U_-$ and $H_{\rm pump}=U_-^\dagger H_{\rm seed}U_- - \mathrm{i}\hbar U_-^\dagger\dot{U}_-$. It makes the second pump coupling static and changes the g_2 diagonal from δ to $\delta+\Omega_{\rm beat}=\omega_{\rm HF}$. The resulting pump Hamiltonian is

$$\frac{H_{\rm pump}}{\hbar}=\omega_{\rm HF}|g_2\rangle\langle g_2|-\sum_e\Delta_e(\boldsymbol{v})|e\rangle\langle e|+\frac{1}{2}\sum_{g,e}\left(\Omega_{ge}^{\rm pump}|e\rangle\langle g|+\Omega_{ge}^{\rm pump*}|g\rangle\langle e|\right), \tag{11}$$

where $\Delta_e(\boldsymbol{v})=\Delta_e-\boldsymbol{k}_{\rm pump}\cdot\boldsymbol{v}$. Spontaneous emission, collisional optical dephasing, and ground-state decoherence enter the same reference Liouvillian. The shared dissipator includes the trace-preserving thermal reload channel $\gamma_t(\rho_{\rm th}\,\mathrm{Tr}\,\rho-\rho)$ with ground-manifold weights 5/12 and 7/12; the density-dependent Rb–Rb term remains a separate pure-dephasing channel. Vectorizing the 4×4 density matrix gives a 16×16 null-space problem; one row is replaced by the trace constraint. Solving that system makes populations and coherences feed back on one another self-consistently.

The diagonal phase in Eq. (10) leaves the declared projector dephasing and thermal reload invariant; phases acquired by individual collapse operators cancel in their Lindblad dissipators. The executable gauge map reconstructs the seeded-frame pump state with g_2-row coherences in $n=-1$, g_2-column coherences in $n=+1$, all remaining entries in $n=0$, and all $|n|\geq2$ harmonics exactly zero. The reconstructed state agrees with the direct $N_F=3$ pump-only Floquet solution to 8.37×10^{-15}. This gauge-parity check applies only to the standard minus Raman branch. The inherited plus-branch finite-seed frame fails this gauge-parity test and the pump-only reference therefore returns `unsupported` rather than mapping it silently.

Equations (9)–(11) are an executable *semianalytic pump-only steady-state reference*: the Liouvillian matrix elements are analytic functions of the parameters and the null space is solved numerically. It remains separate from the current gain tables. The production path instead solves a trace-one, finite-seed density matrix at $N_F\geq3$ and applies an adjacent-order full-scan convergence gate. The generalized audit separately repeats the same finite-seed finite-Floquet construction for $N_F=1,2,3$. The rate/Sylvester calculation in `analysis/squeezing/analytic_reconstruction/theory_final.py` is a third, rate-equation factorization; it is not the source of the corrected Option-A tables. Each solve is exact only within its four-level, Markov, rotating-wave, field, velocity, and Floquet assumptions. In particular, the reference is undepleted, uses the one-dimensional longitudinal Doppler grid by default, and does not recompute the pump state segment by segment. Section 8 adds a separate two-dimensional wrapper around this reference; it does not promote that wrapper into the finite-seed production path.

The finite-seed path takes the probe-seed power already present at the cell as an input. It does not model the apparatus EOM V_π, insertion loss, RF-to- sideband conversion, or sideband impurity. Those quantities belong to the lab-to-cell input map and technical-noise budget, not to the pump-nullspace equation.

5 Validity of the rate/Sylvester factorization

An alternative construction solves saturation rate equations for populations and then solves a Sylvester equation for coherences. Because the resulting coherences do not feed back into the pop-

ulations, that construction is not an exact strong-pump steady state. It may be used only under all of the following assumptions:

- optical coherences can be adiabatically eliminated;
- excited-state interference and coherent population trapping are small;
- pump-induced level dressing is weak compared with relevant separations;
- population redistribution is slow compared with optical dephasing; and
- fitted rates represent the omitted coherent dynamics.

At the dressed-wavevector diagnostic point the quoted pump Rabi rate is comparable to or larger than the excited-state hyperfine splitting ($\Omega_{\mathrm{pump}}/2\pi = 0.708\,\mathrm{GHz}$ versus $\nu_{\mathrm{HF}}^{(e)} = 361.58\,\mathrm{MHz}$), so these assumptions are not automatically satisfied. All population and coherence numbers obtained this way are therefore classified as a controlled approximation only if a separate error study is supplied.

6 Phenomenological rates and truncations

The four-level pump model uses a Zeeman-averaged hyperfine basis. The ground-coherence floor, density-dependent collisional term, and optical self-broadening are phenomenological inputs unless independently measured. At the literature point the engine evaluates

$$\frac{\gamma_{gg}}{2\pi} = 102.882\,\mathrm{kHz}, \qquad \frac{\gamma_{\mathrm{opt}}}{2\pi} = 0.7416\,\mathrm{MHz}, \qquad \frac{\Gamma}{2\pi} = 5.7460\,\mathrm{MHz}, \tag{12}$$

for $N = 2.1496 \times 10^{19}\,\mathrm{m}^{-3}$ at $T = 121\,^{\circ}\mathrm{C}$, where the effective ground-coherence decay is the sum of the 100 kHz trace-preserving transit reset and a 2.882 kHz Rb–Rb pure-dephasing term, and $\gamma_{\mathrm{opt}} = \frac{1}{2}\beta N$ is the added collisional optical half-width. Thus the transit contribution replenishes ground populations rather than merely damping the Raman coherence, while preserving trace exactly. Their provenance and sensitivity are listed in Appendix D. No calculation involving these quantities is labeled exact.

Part III
Weak-field linear response

7 Frequency-dependent response matrix

Let $\mathcal{V}_k(\Omega_{\mathrm{SA}})$ be the perturbation generated by the weak probe or conjugate sideband, and let $\omega_k(\delta, \Omega_{\mathrm{SA}})$ be that field's angular frequency in the static pump frame. With Eqs. (5) and $\dot{\rho} = \mathcal{L}_{\mathrm{pump}}\rho + \mathcal{V}\rho_{\mathrm{ss}}$,

$$\delta\rho_k(\Omega_{\mathrm{SA}}) = -\left(\mathcal{L}_{\mathrm{pump}} + \mathrm{i}\omega_k(\delta, \Omega_{\mathrm{SA}})\right)^{-1}_{\mathrm{Tr}=0} \mathcal{V}_k(\Omega_{\mathrm{SA}})\rho_{\mathrm{ss}}. \tag{13}$$

For the implemented minus-branch Nambu sector,

$$\omega_k^{(-)}(\delta, \Omega_{\mathrm{SA}}) = -\omega_{\mathrm{HF}} + \delta + \Omega_{\mathrm{SA}}. \tag{14}$$

Thus the optical two-photon detuning and RF analysis frequency are independent inputs; changing one never silently changes the other. The inverse acts on the trace-zero subspace. At $\omega_k = 0$ the implementation projects the full stationary subspace instead of applying an ordinary singular inverse. This zero relative resolvent frequency is not generally the same as setting the lab analysis frequency Ω_{SA} to zero. Using the opposite Fourier sign would reverse the sign of the imaginary term; mixing those conventions is not allowed.

Polarization readout gives the full response matrix

$$\begin{pmatrix} \mathcal{P}_p \\ \mathcal{P}_c^* \end{pmatrix} = \epsilon_0 \begin{pmatrix} \chi_{pp}(\Omega_{\mathrm{SA}}) & \chi_{pc}(\Omega_{\mathrm{SA}}) \\ \chi_{cp}^*(\Omega_{\mathrm{SA}}) & \chi_{cc}^*(\Omega_{\mathrm{SA}}) \end{pmatrix} \begin{pmatrix} E_p \\ E_c^* \end{pmatrix}. \tag{15}$$

Thus M_q, the reservoir coupling B, and the diffusion matrix D may all depend on Ω_{SA}. The reference returns the complex 2×2 derivative in Nambu input/output order (p, c^*). It differentiates with respect to angular Rabi frequency, so the response has units of seconds. It supplies the atomic-response ingredient needed for a future $M_q(\Omega_{\mathrm{SA}})$ construction, but it does not construct B or D and is not the production gain solver.

8 Non-collinear Doppler geometry

Choose the pump along z and the probe-seed in the x–z plane:

$$\boldsymbol{k}_{\mathrm{pump}} = k_{\mathrm{pump}} \hat{\boldsymbol{z}}, \qquad \boldsymbol{k}_p = k_p(\sin\theta\, \hat{\boldsymbol{x}} + \cos\theta\, \hat{\boldsymbol{z}}).$$

The laboratory detunings seen by an atom with velocity $\boldsymbol{v} = (v_x, v_y, v_z)$ are therefore

$$\Delta_{\mathrm{eff}} = \Delta - k_{\mathrm{pump}} v_z, \tag{16}$$

$$\delta_{\mathrm{eff}} = \delta + (k_{\mathrm{pump}} - k_p \cos\theta) v_z - k_p \sin\theta\, v_x. \tag{17}$$

The sign of the transverse term follows directly from $\delta_{\mathrm{eff}} = \delta - (\boldsymbol{k}_p - \boldsymbol{k}_{\mathrm{pump}}) \cdot \boldsymbol{v}$. The quantitative average is at least two-dimensional:

$$\langle X \rangle_v = \iint \frac{\mathrm{d}v_z\, \mathrm{d}v_x}{2\pi\sigma_v^2} \exp\left[-\frac{v_z^2 + v_x^2}{2\sigma_v^2}\right] X(\Delta_{\mathrm{eff}}, \delta_{\mathrm{eff}}). \tag{18}$$

The unused v_y projection integrates to one. For $k_{\mathrm{pump}} \simeq k_p = k$,

$$|\boldsymbol{k}_{\mathrm{pump}} - \boldsymbol{k}_p| = 2k \sin(\theta/2) \simeq k\theta, \qquad \frac{\sigma_{\delta,\theta}}{2\pi} \simeq \frac{\theta\sigma_v}{\lambda} = 1.361\,\mathrm{MHz} \tag{19}$$

for $\theta = 0.32°$, $\sigma_v = 193.7\,\mathrm{m/s}$, and $\lambda = 795\,\mathrm{nm}$. These values correspond to the 110 °C estimate. At the 121 °C literature point, $\sigma_v \simeq 196.5\,\mathrm{m/s}$ and the width is about 1.38 MHz. Both are much larger than $\gamma_{gg}/2\pi = 0.103\,\mathrm{MHz}$. The angular term vanishes as $\theta \to 0$; a roughly 2 kHz collinear residual remains because $\omega_{\mathrm{pump}} \neq \omega_p$.

The minus-branch pump-only reference implements Eq. (18) as a tensor Gauss–Legendre quadrature of the truncated Maxwell distribution. Its interface keeps $\Omega_{\mathrm{beat,lab}}$ and Ω_{SA} independent of velocity and applies Eq. (17) only to the atomic detunings. This separation is essential: passing δ_{eff} to the production Floquet table would also move the laboratory beat frequency and would describe a different problem.

At the literature point the analytic rms is 1.380173564 MHz, of which 1.380161171 MHz is transverse and 5.848714 kHz is axial at angle. The 40×40, $5\sigma_v$ quadrature gives 1.380163304 MHz, a -7.43×10^{-4}% difference. Table 3 records the feature-level convergence. These large gains are mean-field diagnostics under the same uncalibrated coupling assumptions as the other gain tables, not experimental predictions.

The accepted successive-grid comparison is $32 \to 40$: the gain changes by 0.00711% and the sampled feature position by 0.05 MHz. Extending the cutoff from $4.5\sigma_v$ to $5\sigma_v$ changes the gain by 0.05215% and does not move the sampled feature. At the declared −8 MHz literature point, the one-dimensional and two-dimensional diagnostics are respectively $G_p = 403.825150$ and 404.304420, a +0.118683% change. The response residual is 2.18×10^{-16} and the trace error is 5.83×10^{-26}.

These quadrature checks apply only to the slow reference: the finite-seed scan remains one-dimensional, the crossing angle is a single value rather than a beam angular distribution, and the pump state is not recomputed segment by segment. No microscopic noise is added.

Table 3: Two-dimensional non-collinear reference convergence. The refined scan spacing is 0.05 MHz; the first three rows use a $5\sigma_v$ cutoff.

Grid or cutoff	velocity pairs	feature $\delta/2\pi$	G_p	change
24×24	576	36.15 MHz	4517.4025	+0.01495% vs 40
32×32	1024	36.00 MHz	4517.0481	+0.00711% vs 40
40×40	1600	35.95 MHz	4516.7271	reference
$4.5\sigma_v$ at 40	1600	35.95 MHz	4519.0828	+0.05215%
$5.0\sigma_v$ at 40	1600	35.95 MHz	4516.7271	reference

9 Floquet harmonics and convergence requirement

If the chosen rotating frame leaves a periodic pump term at Ω_{beat}, expand

$$\rho(t) = \sum_{n=-N_F}^{N_F} \rho_n \mathrm{e}^{-\mathrm{i}n\Omega_{\mathrm{beat}}t}.$$

The finite block equations are

$$(\mathcal{L}_0 + \mathrm{i}n\Omega_{\mathrm{beat}})\,\rho_n + \mathcal{C}_p\rho_{n-1} + \mathcal{C}_m\rho_{n+1} = 0. \tag{20}$$

Trace one is imposed in the $n = 0$ sector and trace zero in every other sector. Before truncation, the same blocks define an infinite-dimensional Floquet–Liouville operator $\mathbb{L}$; its eigenvalues are Floquet exponents, defined modulo $\mathrm{i}\Omega_{\mathrm{beat}}$. When the relevant spectral decomposition exists, the resolvent identities of Sec. 10 then apply on that extended space, with every harmonic sector contributing a frequency replica; a simple-pole sum additionally requires diagonalizability.

For the exact periodic GKSL problem, the one-period propagator is CPTP, so each Floquet multiplier $r_m = \exp(\lambda_m T)$ obeys $|r_m| \le 1$. A finite-N_F projected block matrix is not generally the generator or logarithm of a CPTP map, however; positive real parts may be truncation artifacts, and passing a half-plane or state-positivity test does not establish convergence. The truncation is accepted only after convergence with respect to N_F has been established for the transfer coefficients, their wrapped phases, and the specified optimum δ^*. The production gate uses the dimensionless mixed tolerance $|\Delta T| \le 10^{-10} + 0.01\max(|T_N|, |T_{N-1}|)$ for every transfer coefficient. Reduced susceptibilities have units of time, so their absolute floor is instead 10^{-8} times that component′s full-scan maximum; this is added to the same one-percent local relative term. Derived gains must change by less than one percent, retained wrapped phases by less than 0.5°, and the probe-gain optimum by at most one final-grid interval. A failed condition returns `UNCONVERGED`. The production core and compiled kernel use generalized block continued fractions and are checked for $N_F = 1, 2, 3$ against a separately assembled dense finite block, including trace and normalized-equation residual tests. The diagnostic calculation in Sec. 36 retains its own assembly, reproduces `ref_solver.py`′s diagnostic $N_F = 1$ finite-seed solve, and agrees with direct finite-block solves to 5.4×10^{-13}. Those numbers isolate truncation under the dressed-k diagnostic convention; they are not production Option-A physical predictions. The separate literature-point section uses the current density-matrix assembly and checks only its Maxwell/Q propagation against a literal independent SI reconstruction. The separate pump-only reference implements Eq. (9) for the minus branch and is checked against both its gauge-mapped finite-Floquet state and an independent dense/direct response construction; it is not substituted into either of those finite-seed paths.

10 Exact spectral representation of the resolvent

Numerical matrix inversion evaluates Eq. (13) at each Ω_{SA}. The equivalent pole representation below identifies the modes contributing to the response.

10.1 Eigenbasis of the Liouvillian

Let $\mathcal{L} \equiv \mathcal{L}_{\text{pump}}$ act on the n^2-dimensional vectorized space and let its right and left eigenvectors be

$$\mathcal{L}\,|r_m\rangle = \lambda_m |r_m\rangle, \qquad \langle l_m|\mathcal{L} = \lambda_m \langle l_m|, \qquad \langle l_m|r_{m'}\rangle = \delta_{mm'}, \tag{21}$$

where $\langle\cdot|\cdot\rangle$ is the Hilbert–Schmidt pairing. $\mathcal{L}$ is not Hermitian, so left and right eigenvectors differ and the eigenvalues are complex. Whenever $\mathcal{L}$ is diagonalizable, the resolvent has the exact partial-fraction form

$$\boxed{(\mathcal{L} + \mathrm{i}\Omega)^{-1} = \sum_m \frac{|r_m\rangle\langle l_m|}{\lambda_m + \mathrm{i}\Omega}\,.} \tag{22}$$

Substituting into Eq. (13) and contracting with the Clebsch–Gordan-weighted polarization readout $\langle w_j|$ gives the reduced response as a pole sum,

$$\boxed{\bar{\chi}_{jk}(\Omega) = \sum_m \frac{A_m^{(jk)}}{\lambda_m + \mathrm{i}\Omega}, \qquad A_m^{(jk)} = -\langle w_j|r_m\rangle\langle l_m|\mathcal{V}_k|\rho_{\text{ss}}\rangle.} \tag{23}$$

Equation (23) is an exact similarity transform of the same linear system, under the assumption that $\mathcal{L}$ is diagonalizable.

For a defective eigenvalue, degeneracy alone is not the issue: a Jordan block $J_\lambda = \lambda I + N$ of size p, with $N^p = 0$, contributes

$$(J_\lambda + \mathrm{i}\Omega I)^{-1} = \sum_{r=0}^{p-1} \frac{(-1)^r N^r}{(\lambda + \mathrm{i}\Omega)^{r+1}}. \tag{24}$$

Thus the resolvent identity survives, but the simple Lorentzian pole sum does not: defective blocks generate higher-order poles and should be evaluated by a Jordan-free linear solve or contour/Riesz-projector construction in numerical work.

10.2 The trace-mode residue and the zero-frequency inverse

$\mathcal{L}$ always has the eigenvalue $\lambda_0 = 0$, whose right eigenvector is the steady state ρ_{ss}. Its term in Eq. (22) diverges as $\Omega \to 0$. For a unique stationary state and the traceless commutator drive used here, its trace-mode residue vanishes; the zero-frequency inverse still requires an explicit trace-zero or Drazin construction.

Trace preservation means $\mathrm{Tr}\,\mathcal{L}[X] = 0$ for every X, so the trace functional is the *left* eigenvector belonging to λ_0:

$$\langle l_0| = \mathrm{Tr}(\,\cdot\,). \tag{25}$$

Assume first that the pump Liouvillian has a unique stationary state. The drive is a commutator, $\mathcal{V}_k\rho_{\text{ss}} = -\mathrm{i}[V_k, \rho_{\text{ss}}]$, and is therefore traceless. Hence

$$A_0^{(jk)} \propto \langle l_0|\mathcal{V}_k|\rho_{\text{ss}}\rangle = \mathrm{Tr}\big(-\mathrm{i}[V_k, \rho_{\text{ss}}]\big) = 0 \tag{26}$$

identically, for any V_k and any ρ_{ss}. For $\Omega \neq 0$ the stationary-mode contribution to the contracted response therefore vanishes. At $\Omega = 0$, however, the full inverse does not exist and the formal stationary term is $0/0$; one must retain the trace-zero solve of Eq. (13), omit the stationary subspace explicitly, or use the Drazin inverse. If the Liouvillian has more than one stationary mode, tracelessness alone does not eliminate the residues of every zero mode, and the whole stationary subspace must be projected. Numerically, the trace-mode residue vanishes exactly for the engine′s Liouvillian and drives.

10.3 Lorentzian reading

Write the eigenvalues as

$$\lambda_m = -\Gamma_m + \mathrm{i}\omega_m, \qquad \Gamma_m \geq 0, \tag{27}$$

and absorb the sign into the residue, $B_m \equiv -A_m$. In the diagonalizable simple-pole case, each term of Eq. (23) is then a complex Lorentzian in standard form,

$$\bar{\chi}_{jk}(\Omega) = \sum_m \frac{B_m^{(jk)}}{\Gamma_m - \mathrm{i}(\Omega + \omega_m)}, \tag{28}$$

For an isolated pole with a slowly varying residue, the amplitude-squared profile has HWHM Γ_m and center $\Omega = -\omega_m = -\operatorname{Im}\lambda_m$. A complex residue mixes absorptive and dispersive components, and overlapping poles need not display a peak exactly at that center. Subject to those qualifications, the parametrization assigns a physical interpretation to the numerically computed spectrum of $\mathcal{L}$:

- $\operatorname{Im}\lambda_m$ is a *dressed oscillation frequency*: a generalized Rabi frequency, an Autler–Townes or hyperfine splitting, or a Raman detuning, depending on which coherence dominates $|r_m\rangle$.
- $-\operatorname{Re}\lambda_m$ is the corresponding *power-broadened relaxation rate*. It is not the bare Γ, γ_{gg}, or γ_{opt}; strong driving mixes the channels and shifts the widths.
- The residues A_m determine which poles contribute to a given response element. A pole with small Γ_m contributes a narrow feature only if its residue is not also small.

10.4 Positivity of the pole half-widths, and what it does not prove

If $\mathcal{L}$ has been established to be a Lindblad generator [11], the semigroup $\mathrm{e}^{\mathcal{L}t}$ is completely positive and trace preserving for all $t \geq 0$. Its spectrum therefore satisfies

$$\operatorname{Re}\lambda_m \leq 0 \quad \text{for every } m, \tag{29}$$

with at least one eigenvalue exactly zero, whose right eigenvector is the steady state. Two consequences follow.

First, Eq. (29) places the poles in the closed stable half plane. Finiteness on every real Ω additionally requires a unique relaxing steady state, or an explicit check that every nonstationary mode has $\operatorname{Re}\lambda_m < 0$; complete positivity alone permits peripheral imaginary modes and decoherence-free sectors.

Second, Eq. (29) is guaranteed for an exactly assembled Lindblad generator. A numerical spectral check is therefore an implementation invariant rather than an independent validation of model quality: it can detect assembly or truncation mistakes, but passing it does not validate omitted physics or prove that a phenomenological reservoir is GKSL. The generator Choi test in Sec. 21 addresses that separate issue. A numerical stability check of the specified reduced matrices finds that at both operating points of Table 2 the largest real part found over the velocity grid is at most $+1.7 \times 10^{-6}\,\mathrm{s}^{-1}$, which is the null eigenvalue resolved to better than 5×10^{-14} relative to the scale $\Gamma = 3.6 \times 10^{7}\,\mathrm{s}^{-1}$. Model errors — including the Floquet truncation error of Sec. 36 — must be caught by convergence and comparison tests instead (Sec. 37).

10.5 Effect of the velocity average

Equation (23) holds *per velocity class*. The velocity average of Sec. 8 replaces the discrete pole set by a distribution: each $\lambda_m(\boldsymbol{v})$ traces a curve as $\boldsymbol{v}$ varies, and the averaged response is an integral over that curve rather than a finite sum of Lorentzians. This is the microscopic origin of inhomogeneous broadening; only the simple-pole limit below reduces to a Voigt profile.

The two pole families respond differently because the Doppler shift enters the excited-state diagonal as $\Delta_{\mathrm{eff}} = \Delta - k_{\mathrm{pump}} v_z$:

- **Optical poles** shift by the full one-photon rms scale $k\sigma_v/2\pi = \sigma_v/\lambda$, about 244 MHz at 110 °C and 247 MHz at 121 °C, and are inhomogeneously smeared out.
- **Ground-Raman poles** cancel that common axial shift in the collinear limit. On the one-dimensional axial grid used below, their center therefore stays near zero, while their *width* varies through velocity-dependent optical pumping; light shifts can produce residual center shifts. The separate full non-collinear reference additionally distributes their centers through the transverse term in Eq. (17); it does not produce the full optical smear.

Section 35 quantifies both statements at the literature operating point. The residual non-collinear distribution $\sigma_{\delta,\theta}$ of Eq. (19) convolves with the velocity-dependent Raman response. The separate reference performs this average, but a measured squeezing bandwidth additionally requires the complete microscopic noise propagation and detector transfer.

11 Faddeeva base integral and Voigt identity

For a single velocity projection,

$$\left\langle \frac{1}{x_0 - kv + \mathrm{i}\Gamma_o} \right\rangle_v = -\frac{\mathrm{i}}{k\sigma_v}\sqrt{\frac{\pi}{2}}\, w\!\left(\frac{x_0 + \mathrm{i}\Gamma_o}{\sqrt{2}\,k\sigma_v}\right), \qquad w(z) = \mathrm{e}^{-z^2}\,\mathrm{erfc}(-\mathrm{i}z). \tag{30}$$

With

$$V(x;\sigma,\gamma) = \frac{\mathrm{Re}\, w[(x+\mathrm{i}\gamma)/(\sqrt{2}\sigma)]}{\sqrt{2\pi}\sigma},$$

the dimensionally correct identity is

$$\boxed{\mathrm{Im}\left\langle \frac{1}{x_0 - kv + \mathrm{i}\Gamma_o} \right\rangle_v = -\pi V(x_0; k\sigma_v, \Gamma_o).} \tag{31}$$

There is no additional factor $1/k$. Equation (30) is an exact single-pole integral equal to the velocity average of a *single* term of the pole sum (23) under the assumption that the pole position is linear in v and its residue is constant. Velocity-dependent populations, Raman self-energies, and the two-detuning geometry violate that assumption and generally require multidimensional quadrature or a justified partial-fraction reduction. A one-dimensional Faddeeva evaluation is not a complete treatment of the experimental geometry.

Part IV
Semiclassical field propagation and gain

12 Classical coupled-mode equation

Use the GABES positive-frequency convention $E_j^{(+)} \propto \alpha_j(z)\mathrm{e}^{\mathrm{i}k_{j,z}z - \mathrm{i}\omega_j t}$ and define the physical susceptibility so that $\mathrm{Im}\,\chi_{jj} > 0$ is passive absorption. For $\Delta k_{\mathrm{geom}} = 2k_{\mathrm{pump}} - k_p - k_c$, the symmetric mismatch gauge is $\widetilde{\alpha}_p = \alpha_p \exp(-\mathrm{i}\Delta k_{\mathrm{geom}} z/2)$ and $\widetilde{\alpha}_c^* = \alpha_c^* \exp(+\mathrm{i}\Delta k_{\mathrm{geom}} z/2)$. It gives diagonal phases $(-\mathrm{i}\Delta k_{\mathrm{geom}}/2, +\mathrm{i}\Delta k_{\mathrm{geom}}/2)$ in the two-component basis $(\widetilde{\alpha}_p, \widetilde{\alpha}_c^*)^T$. This fixes the code-level Maxwell convention directly; in particular a passive uncoupled probe obeys $|\alpha_p(L)|^2 = \exp[-k_p\,\mathrm{Im}\,\chi_{pp}L]|\alpha_p(0)|^2$. Direct differentiation then gives

$$\frac{\mathrm{d}}{\mathrm{d}z}\begin{pmatrix}\widetilde{\alpha}_p \\ \widetilde{\alpha}_c^*\end{pmatrix} = M_{\mathrm{cl}}\begin{pmatrix}\widetilde{\alpha}_p \\ \widetilde{\alpha}_c^*\end{pmatrix}, \qquad M_{\mathrm{cl}} = Q^{-1}M_E Q, \tag{32}$$

where the Maxwell electric-field matrix is

$$M_E = \begin{pmatrix} +\frac{\mathrm{i}k_p}{2}\chi_{pp} - \frac{\mathrm{i}\Delta k_{\mathrm{geom}}}{2} & +\frac{\mathrm{i}k_p}{2}\chi_{pc} \\ -\frac{\mathrm{i}k_c}{2}\chi_{cp}^* & -\frac{\mathrm{i}k_c}{2}\chi_{cc}^* + \frac{\mathrm{i}\Delta k_{\mathrm{geom}}}{2}\end{pmatrix}, \qquad Q = \mathrm{diag}\left(\sqrt{\frac{2\hbar\omega_p}{\epsilon_0 c A_p}}, \sqrt{\frac{2\hbar\omega_c}{\epsilon_0 c A_c}}\right). \tag{33}$$

Here

$$\Delta k_{\mathrm{geom}} = 2k_{\mathrm{pump},z} - k_{p,z} - k_{c,z} \tag{34}$$

uses vacuum carrier wave numbers and geometry only. The effective mode areas A_p, A_c must match the collected modes. Equation (32) is the canonical photon-flux rescaling of the electric-field equation; a Rabi-amplitude implementation must equivalently apply Eq. (7).

No double counting of dispersion

Equation (32) is Option A. The real parts of χ_{pp} and χ_{cc} generate dispersive phase exactly once through the diagonal propagation terms. Refractive indices $n_j = 1 + \mathrm{Re}\,\chi_{jj}/2$ are *not* inserted again into Δk_{geom}. A dressed-k calculation that uses both mechanisms is an explicit dispersion-double-counting counterexample used to isolate numerical convergence.

13 Constant-matrix exponential and mean-field gains

For an undepleted pump and z-independent coefficients,

$$\begin{aligned} T_{\mathrm{cl}}(L) &= \mathrm{e}^{M_{\mathrm{cl}}L} \\ &= \mathrm{e}^{sL}\left[\cosh(qL)I + \frac{\sinh(qL)}{q}(M_{\mathrm{cl}} - sI)\right], \end{aligned} \tag{35}$$

$$s = \frac{a+d}{2}, \qquad q = \sqrt{\frac{(a-d)^2}{4} + bc}, \tag{36}$$

where a, b, c, d are the entries of M_{cl}, so that M_{cl} has eigenvalues

$$\mu_{\pm} = s \pm q. \tag{37}$$

Equation (36) needs no eigenvectors. Split off the trace, $M_{\mathrm{cl}} = sI + N$ with N traceless. For a traceless 2×2 matrix the Cayley–Hamilton theorem reads $N^2 = -\det(N)\, I = q^2 I$, so the exponential series collapses into even and odd parts,

$$\mathrm{e}^{NL} = \sum_{k\ \mathrm{even}} \frac{(qL)^k}{k!} I + \sum_{k\ \mathrm{odd}} \frac{(qL)^k}{k!} \frac{N}{q} = \cosh(qL)\, I + \frac{\sinh(qL)}{q} N, \tag{38}$$

and $\mathrm{e}^{M_{\mathrm{cl}}L} = \mathrm{e}^{sL}\mathrm{e}^{NL}$ because sI commutes with N. Since $\sinh(z)/z$ is entire, Eq. (36) is valid verbatim at $q = 0$, where it reads $\mathrm{e}^{sL}(I + LN)$; no limit needs to be taken. The same 2×2 trace–discriminant parametrization reappears in the quantum-noise integral of Sec. 18. Its numerical pair generally differs because $M_q(\Omega)$ need not equal M_{cl}.

The formula is an algebraic identity for constant M_{cl}. Its physical assumptions are:

- undepleted pump;
- fixed density and temperature;
- no longitudinal pump attenuation;
- z-independent overlap and response coefficients; and
- a single collected transverse mode.

If those assumptions fail, use an ordered product or path-ordered exponential.

13.1 Gain convention

Both gains are referred to the incident probe-seed power P_0 and read from a single photon-flux transfer matrix evaluated with probe-seed-only input:

$$\boxed{G_p \equiv \frac{P_p^{\mathrm{out}}}{P_0} = |T_{\mathrm{cl},11}|^2, \qquad G_c \equiv \frac{P_c^{\mathrm{out}}}{P_0} = \frac{\omega_c}{\omega_p}|T_{\mathrm{cl},21}|^2.} \tag{39}$$

This is a **two-gain convention**: G_p and G_c are computed independently. Two gaps must be distinguished:

$$\mathcal{G}_{\text{flux}} \equiv |T_{\text{cl},11}|^2 - |T_{\text{cl},21}|^2 = G_p - \frac{\omega_p}{\omega_c} G_c, \qquad \mathcal{G}_{\text{power}} \equiv G_p - G_c. \tag{40}$$

Both are outputs rather than inputs. Canonical commutator preservation constrains $\mathcal{G}_{\text{flux}}$, which equals one in the closed lossless parametric limit of Eq. (43). The power gap is numerically close because $\omega_p \simeq \omega_c$, but it is not the canonical invariant.

The distinction matters because a near-degenerate single-gain power convention is also in use. The exact photon-flux relation is

$$P_p = G\, P_0, \qquad P_c = \frac{\omega_c}{\omega_p}(G-1)\, P_0. \tag{41}$$

The commonly quoted $P_c = (G-1)P_0$ form additionally uses $\omega_c/\omega_p \simeq 1$. In either form the conjugate photon-flux gain has been imposed rather than computed, and $\mathcal{G}_{\text{flux}} = 1$ by construction. Equation (41) is the ideal-parametric special case of Eq. (39). The mapping between the two conventions is

$$G_p \;\longleftrightarrow\; G, \qquad G_c \;\longleftrightarrow\; \frac{\omega_c}{\omega_p}(G-1), \tag{42}$$

valid only where $\mathcal{G}_{\text{flux}} = 1$. Two consequences follow. A conjugate-to-probe-seed power ratio quoted under Eq. (41) is not an independently calibrated G_c, so it must not be compared with the computed G_c as though it were. The photon-flux gap also cannot serve as a validation target against a data set in which it was assumed.

The quantities in Eq. (39) are explicitly **semiclassical mean-field gains**. They do not by themselves determine a quantum covariance.

14 Classical photon-flux identity

For $u' = au + bv$ and $v' = cu + dv$,

$$\frac{\mathrm{d}}{\mathrm{d}z}(|u|^2 - |v|^2) = 2\,\mathrm{Re}(a)|u|^2 - 2\,\mathrm{Re}(d)|v|^2 + 2\,\mathrm{Re}[(b-c^*)u^*v]. \tag{43}$$

In a closed lossless parametric model, $\mathrm{Re}\, a = \mathrm{Re}\, d = 0$ and $b = c^*$, so $\mathcal{G}_{\text{flux}} = 1$. Equation (43) is a property of the classical transfer matrix. If $\mathcal{G}_{\text{flux}} \neq 1$, it is evidence that a two-operator quantum map needs additional channels; it is not permission to use a noncanonical Bogoliubov map. Because the two-gain convention computes $\mathcal{G}_{\text{flux}}$ rather than imposing it, this test is available at every scan point.

Part V
Quantum fluctuation propagation

15 Quantum-Langevin propagation

Define the two-mode sideband vector in the Bogoliubov (phase-conjugate) pairing that the four-wave process couples,

$$\hat{\boldsymbol{A}}(z,\Omega) = \begin{pmatrix} \delta\hat{a}_p(z,\Omega) \\ \delta\hat{a}_c^\dagger(z,-\Omega) \end{pmatrix}, \qquad [\hat{\boldsymbol{A}}(z,\Omega), \hat{\boldsymbol{A}}^\dagger(z,\Omega')] = 2\pi J\delta(\Omega-\Omega'), \quad J = \mathrm{diag}(1,-1). \tag{44}$$

The metric J is the one preserved by a two-mode Bogoliubov transformation; a lossless parametric amplifier acts on $\hat{\boldsymbol{A}}$ by an element of $U(1,1)$ (and $SU(1,1)$ after removing an overall phase), and the

requirement $TJT^\dagger = J$ used below is exactly that condition. Equivalently, in the quadrature basis of Eq. (52) it is the symplectic condition. The distributed fluctuation equation is

$$\frac{\partial}{\partial z}\hat{\boldsymbol{A}}(z,\Omega_{\rm SA}) = M_q(z,\Omega_{\rm SA})\hat{\boldsymbol{A}}(z,\Omega_{\rm SA}) + B(z,\Omega_{\rm SA})\hat{\boldsymbol{F}}(z,\Omega_{\rm SA}). \tag{45}$$

Its formal solution is

$$T(z_2,z_1;\Omega) = \mathcal{P}\exp\left[\int_{z_1}^{z_2} M_q(z,\Omega)\mathrm{d}z\right], \tag{46}$$

$$\hat{\boldsymbol{A}}_{\rm out}(\Omega) = T(L,0;\Omega)\hat{\boldsymbol{A}}_{\rm in}(\Omega) + \int_0^L T(L,z;\Omega)B(z,\Omega)\hat{\boldsymbol{F}}(z,\Omega)\mathrm{d}z. \tag{47}$$

For local Markov reservoirs,

$$[\hat{\boldsymbol{F}}(z,\Omega),\hat{\boldsymbol{F}}^\dagger(z',\Omega')] = 2\pi J_F\delta(z-z')\delta(\Omega-\Omega').$$

16 Preservation of the commutation relations

The output operators must obey the same commutation relations as the input operators. With the reservoir contribution included this requires

$$\boxed{TJT^\dagger + \int_0^L T(L,z)BJ_FB^\dagger T^\dagger(L,z)\mathrm{d}z = J,} \tag{48}$$

which is the statement that gain and loss must each be accompanied by their own Langevin noise. A sufficient local condition is

$$\boxed{M_qJ + JM_q^\dagger + BJ_FB^\dagger = 0.} \tag{49}$$

Define the Hermitian commutator-defect matrix

$$K \equiv -\left(M_qJ + JM_q^\dagger\right) = BJ_FB^\dagger. \tag{50}$$

The signs of its eigenvalues classify the reservoir channels the drift requires: a positive eigenvalue of K admits an annihilation-noise channel with $J_F = +1$, while a negative eigenvalue requires a creation-noise channel with $J_F = -1$. These signs specify reservoir operator ordering, not atomic population inversion. Both channels can occur in a parametric amplifier; the sign pattern is a *channel classification*, not a validity test. Because K is defined through the canonical metric J, it is sensitive to the normalization of Eq. (7) and must be recomputed whenever the collected mode areas or Rabi-to-field factors change.

17 Covariance and atomic diffusion

Define reservoir symmetrized correlations by

$$\frac{1}{2}\left\langle\{\hat{\boldsymbol{F}}(z,\Omega),\hat{\boldsymbol{F}}^\dagger(z',\Omega')\}\right\rangle = 2\pi N_{\rm res}(z,\Omega)\delta(z-z')\delta(\Omega-\Omega'), \qquad D = BN_{\rm res}B^\dagger.$$

Then

$$\boxed{V_{\rm out}(\Omega) = T(\Omega)V_{\rm in}(\Omega)T^\dagger(\Omega) + W(L;\Omega), \qquad W(L;\Omega) \equiv \int_0^L T(L,z;\Omega)D(z,\Omega)T^\dagger(L,z;\Omega)\mathrm{d}z.} \tag{51}$$

Equation (51) is one 2×2 two-mode block. Before applying the detector model, solve its companion block and assemble

$$\hat{\boldsymbol{\Xi}}(\Omega) = \left(\hat{a}_p(\Omega),\hat{a}_c(\Omega),\hat{a}_p^\dagger(-\Omega),\hat{a}_c^\dagger(-\Omega)\right)^T$$

with spectral covariance $V_\Xi(\Omega)$. The quadrature-sideband covariance used below is

$$V_R(\Omega) = C V_\Xi(\Omega) C^\dagger, \qquad C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ -\mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -\mathrm{i} & 0 & \mathrm{i} \end{pmatrix}, \tag{52}$$

corresponding to $\hat{\boldsymbol{R}} = (X_p, Y_p, X_c, Y_c)^T$. This explicit assembly prevents the 2×2 two-mode covariance from being mixed dimensionally with the 4×4 detector matrices.

The commutator condition fixes antisymmetric reservoir correlations, not the symmetrized diffusion that determines squeezing. A physical $D(z, \Omega_{\mathrm{SA}})$ must be derived from the pumped atomic Lindblad model, generalized Einstein relations, or an independently justified reservoir model. Independent velocity classes add as independent diffusion powers; one must not square a velocity-averaged noise amplitude.

18 Closed-form evaluation of the distributed-noise integral

The distributed-noise integral $W(L;\Omega)$ in Eq. (51) has a closed form for a z-independent drift. Its 2×2 trace–discriminant structure parallels the mean-field transfer matrix, but it is formed from $M_q(\Omega)$ itself.

18.1 Differential Lyapunov form

For constant M_q the propagator depends only on the separation, $T(L, z) = \mathrm{e}^{M_q(L-z)}$, so substituting $u = L - z$ gives

$$W(L) = \int_0^L \mathrm{e}^{M_q u}\, D\, \mathrm{e}^{M_q^\dagger u}\, \mathrm{d}u. \tag{53}$$

Introduce the *Lyapunov superoperator*, a linear map on matrices,

$$\mathcal{M}[X] \equiv M_q X + X M_q^\dagger, \qquad \text{so that} \qquad \mathrm{e}^{u\mathcal{M}}[D] = \mathrm{e}^{M_q u} D\, \mathrm{e}^{M_q^\dagger u}. \tag{54}$$

The second identity holds because left and right multiplication commute. Equation (53) is then simply $\int_0^L \mathrm{e}^{u\mathcal{M}}[D]\mathrm{d}u$, and differentiating with respect to the upper limit gives the differential Lyapunov equation

$$\boxed{\frac{\mathrm{d}W}{\mathrm{d}L} = \mathcal{M}[W] + D, \qquad W(0) = 0.} \tag{55}$$

W is Hermitian for Hermitian D, because $\mathcal{M}$ preserves Hermiticity, and positive semidefinite for positive semidefinite D, because the integrand of Eq. (53) is a congruence of D. These invariants detect numerical errors: a computed W that violates either, beyond rounding error, is inconsistent with the stated assumptions.

18.2 Solution

Equation (55) is an inhomogeneous linear equation with constant coefficients, so its solution is the elementary quadrature of $\mathrm{e}^{u\mathcal{M}}$:

$$\boxed{W(L) = L\, \varphi_1(L\mathcal{M})\,[D], \qquad \varphi_1(z) \equiv \frac{\mathrm{e}^z - 1}{z} = \sum_{k\geq 0} \frac{z^k}{(k+1)!}\,.} \tag{56}$$

The coordinate-free form remains regular because φ_1 is an *entire* function — the series on the right converges everywhere and $\varphi_1(0) = 1$ — so Eq. (56) needs no special case, no removable-pole treatment, and no genericity assumption on M_q. The mean-field transfer matrix of Sec. 13 is built from $\sinh(z)/z = \sum_{k\geq 0} z^{2k}/(2k+1)!$, also entire and also equal to 1 at the origin, so neither layer

has a singularity at a coalescing eigenvalue. An eigenbasis can therefore fail even when the finite-L matrix function is regular. This numerical fact does not make an exceptional point unphysical: eigenmode coalescence, non-normal amplification, and enhanced parameter sensitivity can remain observable properties of the drift.

18.3 Spectral resolution in trace–discriminant form

To identify *which* exponents govern growth and decay, resolve Eq. (56) in the eigenbasis. If $M_q = S\Lambda S^{-1}$ with $\Lambda = \mathrm{diag}(\mu_1, \mu_2)$, the eigenmatrices of $\mathcal{M}$ are $S_{\cdot m} S^{\dagger}_{n\cdot}$ and

$$\mathrm{spec}\,\mathcal{M} = \{\mu_m + \mu_n^*\}, \tag{57}$$

so with $\widetilde{D} = S^{-1} D S^{-\dagger}$,

$$W(L) = S\,\widetilde{W}(L)\,S^{\dagger}, \qquad \widetilde{W}_{mn}(L) = \widetilde{D}_{mn}\,L\,\varphi_1[(\mu_m + \mu_n^*)L] = \widetilde{D}_{mn}\,\frac{\mathrm{e}^{(\mu_m+\mu_n^*)L} - 1}{\mu_m + \mu_n^*}. \tag{58}$$

For the 2×2 two-mode block, define at each analysis frequency

$$s_q \equiv \frac{\mathrm{tr}\,M_q}{2}, \qquad q_q^2 \equiv \frac{(a_q - d_q)^2}{4} + b_q c_q, \qquad \mu_{\pm} = s_q \pm q_q, \tag{59}$$

where a_q, b_q, c_q, d_q are the entries of $M_q(\Omega)$. Then Eq. (57) contains four generically distinct values,

$$\mu_{\pm} + \mu_{\pm}^* = 2\,\mathrm{Re}(s_q \pm q_q), \qquad \mu_{\pm} + \mu_{\mp}^* = 2\,\mathrm{Re}\,s_q \pm 2\mathrm{i}\,\mathrm{Im}\,q_q. \tag{60}$$

Within the fluctuation problem, $\mathrm{Re}(s_q \pm q_q)$ set how fast each eigenchannel amplifies and accumulates diffusion, while $\mathrm{Im}\,q_q$ sets the interference between channels. These equal the classical (s, q) only when $M_q(\Omega) = M_{\mathrm{cl}}$. If $\mathrm{Re}(\mu_m + \mu_n^*) < 0$ for every pair, $W(L)$ saturates as $L \to \infty$ at the algebraic Lyapunov solution $\widetilde{W}_{mn} = -\widetilde{D}_{mn}/(\mu_m + \mu_n^*)$; exponential amplification at large propagation length requires an eigenvalue with positive real part, and diffusion grows exponentially when it excites the corresponding mode. Finite-length gain can also arise from a non-normal drift whose eigenvalues all have nonpositive real parts.

18.4 Numerically stable evaluation

Equation (58) supports spectral *interpretation*, but numerical evaluation requires S to be invertible and well conditioned. For a non-normal parametric drift M_q, the limit $q_q \to 0$ can approach a nontrivial defective matrix: its eigenvectors coalesce and S^{-1} diverges. A scalar-matrix degeneracy does not require eigenvector coalescence. The underlying $W(L)$ remains finite there, by Eq. (56); it is the coordinate system that fails, not the solution. Direct testing bears this out: the eigenbasis form loses all accuracy for a defective drift, while the coordinate-free form agrees with direct quadrature to 2×10^{-15} at the same point (Table 8).

Any stable evaluation of φ_1 therefore suffices. A convenient choice is the block exponential [10]: assemble

$$Z = \begin{pmatrix} -M_q & D \\ 0 & M_q^{\dagger} \end{pmatrix}, \qquad \mathrm{e}^{ZL} = \begin{pmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{pmatrix}, \tag{61}$$

from which

$$\boxed{W(L) = F_{22}^{\dagger} F_{12}\,.} \tag{62}$$

Equation (62) is Eq. (56) evaluated by a single 4×4 matrix exponential, without an eigenvector inverse, avoiding the conditioning problem above.

18.5 Longitudinally varying drift

When M_q and D vary with z — through pump depletion, density gradients, or the Gaussian overlap profile — no single closed form exists. If M_k and D_k are the actual local coefficients, the segmented composition is exact within each piecewise-constant segment. Writing $T_k = \mathrm{e}^{M_k \mathrm{d}z}$ and $W_k = W(\mathrm{d}z; M_k, D_k)$ for segment k, the covariance accumulates as

$$V^{(k+1)} = T_k V^{(k)} T_k^\dagger + W_k, \tag{63}$$

with each W_k evaluated by Eq. (62). This is the noise-carrying counterpart of the segmented mean-field propagation and introduces no approximation beyond piecewise-constant coefficients.

18.6 Static independent-vacuum completion fixture

For testing only, any constant drift may be given this particular local commutator completion. Using K from Eq. (50), let

$$K = U\Lambda_K U^\dagger, \quad B_{\mathrm{vac}} = U|\Lambda_K|^{1/2}, \quad J_F = \mathrm{diag}(\mathrm{sgn}\,\Lambda_K), \quad D_{\mathrm{vac}} = \tfrac{1}{2} B_{\mathrm{vac}} B_{\mathrm{vac}}^\dagger. \tag{64}$$

This eigenfactor closes each channel of K with one independent vacuum port and preserves the commutator by construction. The subscript ″vac″ denotes this chosen independent-vacuum algebraic fixture; it is not a universal lower bound over microscopic atomic reservoirs. In the special phase-insensitive amplifier setting, an analogous independent-vacuum construction gives the familiar Caves added-noise bound [5], but an arbitrary dissipative two-mode drift need not satisfy those assumptions. The fixture is not a derivation of atomic spontaneous-emission noise, and its symmetrized diffusion cannot be promoted to a physical bound without a microscopic model or data. When M_q and the diagnostic detector model are frequency independent, D_{vac} is also frequency independent, and feeding it through Eqs. (56) and (114) yields a flat S_-; this checks covariance propagation and readout, but predicts no bandwidth.

19 What a static completion can and cannot determine

The final in-cell observable is a spectrum $S_-^{\mathrm{out}}(\Omega_{\mathrm{SA}})$. The static independent-vacuum fixture above is paired with a static detector and gives a flat curve. It cannot determine atomic frequency-dependent noise, group delay, a measured squeezing bandwidth, low-frequency technical noise, the apparatus-specific 2 MHz laser-source peak or detector transfer. Microscopic reduced-model spectra are instead constructed from the same atomic generator and its jumps in Part VI. Their additional thermal-ensemble, nonlocal-field and measured-input requirements remain explicit.

Part VI
Microscopic quantum response and moving-atom transport

20 The precise object being validated

Model validation requires three tests: whether its equations define an admissible quantum process, whether its numerical solution approximates those equations, and whether its chosen degrees of freedom, reservoirs, boundary conditions and input parameters describe the apparatus. Passing one test does not imply the next. The microscopic construction below supplies frequency-dependent drift, ordered noise and photocurrent spectra for explicitly specified reduced models. It does not identify the phenomenological mean-field gains elsewhere in this document with a complete quantum channel.

In particular, a commutator-preserving vacuum completion of an arbitrary drift and diffusion derived from atomic jumps are different physical models.

The local stationary four-level model, periodically driven four-level model, finite spatial mode projection, ballistic single-atom trajectory and thermal inflow ensemble each have a different closure assumption, specified below. Internal QRT and complete-positivity tests constrain the chosen model; they do not validate an experimental squeezing value. The reported thermal ensemble is not converged, and nonlocal propagation through the cell is not yet an established apparatus calculation.

21 Atomic drift and diffusion from one GKSL generator

Write the Hamiltonian in angular-frequency units, $h = H/\hbar$, and put rates inside collapse operators L_r, with units $\mathrm{s}^{-1/2}$:

$$\mathcal{L}\rho = -\mathrm{i}[h,\rho] + \sum_r \left(L_r \rho L_r^\dagger - \tfrac{1}{2}\{L_r^\dagger L_r, \rho\} \right). \tag{65}$$

Choose a complete Hermitian traceless basis F_i satisfying $\operatorname{Tr} F_i F_j = \delta_{ij}$, $i = 1, \ldots, d^2-1$. If $\mu_i = \operatorname{Tr} \rho F_i$, the adjoint equation is exactly affine on this basis:

$$\dot{\mu} = A\mu + b, \qquad A_{ij} = \operatorname{Tr}[F_i \mathcal{L}(F_j)], \quad b_i = \operatorname{Tr}[F_i \mathcal{L}(I/d)]. \tag{66}$$

The real matrix A has units s^{-1}. This representation is exact for one finite atom; it does not assert that the atom has a Gaussian state. For centered operators $\delta F_i = F_i - \mu_i I$, define the ordered covariance $C_{ij} = \operatorname{Tr} \rho\, \delta F_i \delta F_j$. The adjoint product rule gives

$$D_{ij}^{(r)} = \operatorname{Tr} \rho\, [L_r^\dagger, F_i][F_j, L_r], \qquad D = \sum_r D^{(r)}, \tag{67}$$

$$\dot{C} = AC + CA^T + D. \tag{68}$$

The Hamiltonian terms cancel from the product-rule defect. Each $D^{(r)}$ is a Gram matrix: if $Q_i = [F_i, L_r]$, then $D_{ij}^{(r)} = \operatorname{Tr} \rho Q_i^\dagger Q_j$. Hence $D^{(r)} \succeq 0$ for $\rho \succeq 0$. Atomic D has units s^{-1} and must not be confused with spatial field diffusion, whose units are m^{-1}. For a stationary state,

$$AC + CA^T + D = 0, \qquad AK + KA^T + D - D^T = 0, \quad K = C - C^T. \tag{69}$$

The matrix K is the expectation of atomic commutators and depends on the state. It is not the constant field commutator J.

For a mixing stationary state, let $R(\Omega) = (-\mathrm{i}\Omega I - A)^{-1}$. With the Fourier convention of Eq. (5), the ordered spectra are

$$S^{>}(\Omega) = RDR^\dagger, \quad S^{<}(\Omega) = RD^T R^\dagger = S^{>}(-\Omega)^T, \quad S^{\mathrm{sym}} = \tfrac{1}{2}(S^{>} + S^{<}). \tag{70}$$

An independent full-density-matrix quantum regression calculation instead propagates

$$C_{ij}(t) = \operatorname{Tr}\left[\delta F_i\, e^{\mathcal{L}t}(\delta F_j \rho_{\mathrm{ss}})\right], \quad t \geq 0, \tag{71}$$

and adds the negative-time adjoint to its one-sided transform. Equality with Eq. (70) tests both source ordering and resolvent signs. An independent probe of 30 random GKSL generators in dimensions $2, 3, 4$, each with four complex jumps and five RF frequencies, found a maximum relative ordered-spectrum discrepancy 9.99×10^{-16} and maximum normalized commutator identity residual 6.58×10^{-16}. These 150 comparisons validate the finite-dimensional identities, not an alkali-vapor reservoir. If a stationary state is nonunique or has nondecaying peripheral modes, projecting those modes out discards elastic or preparation-dependent terms. The stationary continuous-spectrum formula must then be supplemented by those terms; a pseudoinverse alone is not a physical prescription.

21.1 Reservoir admissibility and the meaning of a transit reset

For $\rho_{\rm th} = \sum_i p_i |i\rangle\langle i|$, the operators $L_{ij} = \sqrt{\gamma p_i}|i\rangle\langle j|$ give

$$\mathcal{L}_{\rm reset}\rho = \gamma(\rho_{\rm th}\,\mathrm{Tr}\,\rho - \rho). \tag{72}$$

This is a trace-preserving thermal reload channel. It is a Markov replacement model, not a derivation of finite-flight transit times or a beam boundary condition. In a ballistic trajectory model, applying it again merely because the atom crosses numerical segments would reset the same atom repeatedly.

Trace preservation, Hermiticity preservation and conditional complete positivity of the generator must be checked separately. In the Choi test, project the generator Choi matrix onto the space orthogonal to the maximally entangled vector and require positive semidefiniteness there. Positivity of a particular stationary density matrix does not imply this property. A coherence-only transit-damping variant has a projected-Choi eigenvalue about -0.00607 in its normalized generator audit and a normalized Choi eigenvalue about -3.08×10^{-4} after 1 ns. This counterexample excludes that variant as a general GKSL reservoir even when its computed steady state looks admissible. Passing a radiative or selected thermal-reset model does not validate every phenomenological collision law.

22 Reciprocal SI normalization and hyperfine reduction

For a Gaussian beam with $1/e^2$ intensity radius w,

$$I_0 = \frac{2P}{\pi w^2}, \quad E_0 = \sqrt{\frac{2I_0}{\epsilon_0 c}}, \quad \Omega = \frac{dE_0}{\hbar}, \quad h_{eg} = \frac{\Omega}{2}. \tag{73}$$

The same signed dipole must enter the drive, emitted polarization and radiative normalization. In the scalar D1 convention used here,

$$d_\Gamma^2 = \frac{3\pi\epsilon_0\hbar c^3\Gamma}{\omega_{D1}^3}. \tag{74}$$

It gives $d_\Gamma = 2.5367896004\times10^{-29}\,\mathrm{C\,m}$ for the declared constants. A stored value $2.5377\times10^{-29}\,\mathrm{C\,m}$ implies a linewidth 0.0717886% larger. There is no additional universal $\sqrt{2}$ multiplier: a change of reduced-matrix-element convention requires changing the associated angular factors too. A diagnostic pump/weak-field comparison gives implicit dipoles $1.4646827595\times10^{-29}$ and $7.3257088906\times10^{-30}\,\mathrm{C\,m}$, a ratio 1.9993734141. One cannot repair such a mismatch by fitting output gain alone.

Let $S_{FF'}$ be hyperfine strengths normalized within a ground manifold. For an unpolarized manifold with population p_F, weak absorption in a fixed linear polarization has the population-weighted factor

$$\frac{3C_F^2 p_F}{2F+1} = \frac{p_F S_{FF'}}{3}, \qquad 3C_F^2 = \frac{(2F+1)S_{FF'}}{3}. \tag{75}$$

For the four D1 branches $(2,2),(2,3),(3,2),(3,3)$, the required per-manifold factors $S_{FF'}/3$ are $(2,7,5,4)/27$. The alternative common-factor prescription $3C_F^2/12$ gives $(10,35,35,28)/324$, with ratios 5/12, 5/12, 7/12, 7/12 to those factors. Thus no single fitted scalar can replace both manifold denominators 5 and 7 when p_2, p_3 evolve independently. For a thermal distribution $p_F = (2F+1)/12$, the exact weak absorption factor is $3C_F^2/12$ *after* including population once. Multiplying it by p_F again is incorrect.

A four-state surrogate may use $d_{FF'} = d_\Gamma\sqrt{S_{FF'}/3}$ with trace-one manifold populations. It reproduces this weak absorption average, but not the signs, coherences, polarization selection or optical pumping of the full Zeeman Hamiltonian under a strong drive. The residual $\ell_s = 0.74$ in the separate phenomenological model is therefore an assumption, not a microscopic line-strength determination.

The optical carrier must also refer to the same transition as the detuning. For the specified ground and excited splittings, the transition anchor differs from the fine-structure centroid by $(7\omega_{\rm hf,g} + 5\omega_{\rm hf,e})/12$, or 1.921502256083 GHz in ordinary frequency. Rounding the data tables can leave a 1 kHz splitting discrepancy. In one weak-gain fixture a carrier-only change moves G_p from 1.0517029257 to 1.0517034724; its smallness in that fixture does not justify mixing carrier conventions in a derivation.

23 Eliminating the atom without guessing field noise

Let n denote number density, $A_\perp$ the collected area and $\lambda = nA_\perp$ the line density. Write the field amplitude as $E_j = Q_j a_j$, $Q_j = \sqrt{2\hbar\omega_j/(\epsilon_0 c A_\perp)}$ and define $g_j = d_j Q_j/(2\hbar)$. Then g_j has units $\mathrm{s}^{-1/2}$. For the Nambu vector $b = (a_p, a_c^\dagger)^T$, choose atomic operators O_j consistently with its creation/annihilation ordering. A reciprocal construction has, with signs fixed by that choice,

$$W_{ji} = \mathrm{Tr}(O_j F_i), \quad B_{ij} = -\mathrm{i} g_j \,\mathrm{Tr}\{F_i[O_j^\dagger, \rho_{\mathrm{ss}}]\}, \quad C_0 = -\mathrm{i} J\,\mathrm{diag}(g_j) W. \tag{76}$$

In a stationary local medium the linearized equations and the slice-averaged atomic source normalization are

$$\partial_t \delta F = A\delta F + Bb + \xi, \quad \partial_z b = \lambda C_0 \delta F + K_{\mathrm{geom}} b, \quad \langle \xi(z,t)\xi(z',t')^T\rangle = \frac{D}{\lambda}\delta(z-z')\delta(t-t'). \tag{77}$$

Eliminating the atom at the appropriate rotating-frame frequency gives

$$M(\Omega) = \lambda C_0 R(\Omega) B + K_{\mathrm{geom}}, \tag{78}$$

$$D_f^{>}(\Omega) = \lambda C_0 R(\Omega) D R(\Omega)^\dagger C_0^\dagger, \qquad D_f^{<}(\Omega) = \lambda C_0 R(\Omega) D^T R(\Omega)^\dagger C_0^\dagger. \tag{79}$$

The conjugate-frequency sectors must be aligned before assembling the physical sideband matrix. Both drift and field diffusion are linear in density. For one common area, $\lambda g_i g_j$ is independent of that area; the coherent seed amplitude still scales as $\sqrt{P/\hbar\omega}$. Independent velocity classes contribute weighted drift and weighted covariance, not the square of an averaged noise amplitude. Every bath must appear once; radiation already represented by an explicitly retained field cannot also be counted as an independent unobserved reservoir without a consistent mode partition.

The local approximation assumes short-range independent atomic fluctuations at different z. It is unsuitable for the same ballistic atom observed in successive illuminated regions. Sections 27–28 replace its spatial delta function by an explicit two-position covariance.

24 Physical sidebands, complete positivity and finite RF modes

At a strictly positive analysis frequency the physical annihilation modes are $(a_{p,+}, a_{c,-}, a_{p,-}, a_{c,+})$, with eight real quadratures. A complex 2×2 Nambu covariance is only one sector and is not by itself a two-oscillator covariance to which an arbitrary real symplectic test may be applied. Given a sector transfer T, its Bogoliubov matrices are $U = \mathrm{diag}(T_{00}, T_{11}^*)$ and $V_{01} = T_{01}$, $V_{10} = T_{10}^*$, with the other entries zero. In an (X_1, X_2, P_1, P_2) ordering its real transfer is

$$X = \begin{pmatrix} \mathrm{Re}(U+V) & -\mathrm{Im}(U-V) \\ \mathrm{Im}(U+V) & \mathrm{Re}(U-V) \end{pmatrix}. \tag{80}$$

Permute it into the declared quadrature ordering and include both RF sectors. If $H = (N^{>} + N^{<})/2$ is the integrated sector noise, its anomalous off-diagonal entry $m = H_{01}$ contributes the real cross block $\left(\begin{smallmatrix} \mathrm{Re}\, m & \mathrm{Im}\, m \\ \mathrm{Im}\, m & -\mathrm{Re}\, m \end{smallmatrix}\right)$; the diagonal blocks are $H_{00} I_2, H_{11} I_2$.

For canonical quadratures with vacuum $I/2$ and $J_R = \bigoplus_j \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the Gaussian channel and state conditions are respectively

$$Y + \frac{\mathrm{i}}{2}(J_R - X J_R X^T) \succeq 0, \qquad V + \frac{\mathrm{i}}{2} J_R \succeq 0. \tag{81}$$

These are distinct from merely requiring $Y \succeq 0$, and from the Nambu commutator identity. The channel and state conditions in Ref. [13] are expressed here in the declared quadrature normalization.

A finite measured RF mode is $a_h = \int \mathrm{d}\Omega\, h(\Omega)^* a(\Omega)/(2\pi)$ with $\int \mathrm{d}\Omega\, |h|^2/(2\pi) = 1$. A top-hat interval of width B in Hz therefore has $|h| = 1/\sqrt{B}$. Its time profile has sinc tails and is not

a compact time gate. Tracing orthogonal vacuum modes after frequency-dependent transfer adds more than the average Y:

$$\overline{X} = \sum_k p_k X_k, \qquad Y_{\text{eff}} = \sum_k p_k Y_k + \tfrac{1}{2}\sum_k p_k (X_k - \overline{X})(X_k - \overline{X})^T, \quad \sum_k p_k = 1. \tag{82}$$

For equally weighted rotations by $\pm\theta$, the retained mode has $\overline{X} = \cos\theta I$, $Y_{\text{eff}} = \sin^2\theta I/2$. Omitting the second term would turn a passive physical process into a CP violation. Equation (82) assumes the discarded input modes are vacuum and initially independent of the retained mode; a nonvacuum multimode input requires its full covariance.

25 Finite coherent seeds and periodic quantum regression

A finite coherent seed makes the reference atom periodic. In a declared signed frequency convention write $h(t) = h_0 + Ve^{-\mathrm{i}\nu t} + V^\dagger e^{\mathrm{i}\nu t}$, with $\nu = -\omega_{\text{hf}} + \delta$ for the chosen Raman branch, and expand $\rho(t) = \sum_q \rho_q e^{-\mathrm{i}q\nu t}$. The diffusion in Eq. (67) is evaluated on $\rho(t)$, not just ρ_0. The exact time-dependent identity is still $\dot{C} = A(t)C + CA(t)^T + D(t)$. For harmonic indices h, k, the lifted matrices are

$$\mathbb{A}_{hk} = A_{h-k} + \mathrm{i}h\nu\delta_{hk} I, \quad \mathbb{D}_{hk} = D_{h-k}, \quad \mathbb{R}(\Omega) = (-\mathrm{i}\Omega I - \mathbb{A})^{-1}. \tag{83}$$

The ordered spectrum is $\mathbb{R}\mathbb{D}\mathbb{R}^\dagger$. To form the lesser source, transpose the *atomic operator indices* in each Fourier coefficient; transposing the entire harmonic matrix changes the harmonic ordering and is generally wrong. A shift of time origin rotates cross-harmonic blocks covariantly rather than deleting them.

The density-state cutoff and response cutoff are separate approximation parameters. Check interior physical blocks under enlargement of both; positivity alone cannot certify harmonic convergence. The exact one-period propagator is CPTP and a mixing periodic solution has spectral radius below one on the traceless subspace. A finite harmonic matrix need not itself be a GKSL generator. An independent density-matrix period reference uses

$$Q(\Omega) = \int_0^T \mathrm{d}s\, e^{\mathrm{i}\Omega s}\mathcal{U}(t_0 + s, t_0), \quad \int_0^\infty \mathrm{d}t\, e^{\mathrm{i}\Omega t}\mathcal{U}(t_0 + t, t_0) = Q(\Omega)[I - e^{\mathrm{i}\Omega T}\mathcal{P}(t_0)]^{-1}, \tag{84}$$

with trace bordering/projection on the decaying subspace. This reference does not obtain its noise by copying the Floquet diffusion matrix.

For finite propagating mean fields β, the mean equation is nonlinear, $\beta' = F(\beta, \beta^*)$. Its zero-frequency fluctuation transfer is the tangent map of this equation. In general the mean output is not obtained by applying that tangent transfer to the input mean. Holding the pump prescribed also remains different from solving three-field pump depletion.

For the reciprocal four-state RMS-dipole fixture with $n = 10^{18}\,\mathrm{m}^{-3}$, $A_\perp = 1.2 \times 10^{-7}\,\mathrm{m}^2$, $L = 12.5\,\mathrm{mm}$, pump $0.6\,\mathrm{W}$, $w = 530\,\mu\mathrm{m}$, $\Delta/(2\pi) = 0.9\,\mathrm{GHz}$, $\delta/(2\pi) = -8\,\mathrm{MHz}$, reset rate $2\pi\,100\,\mathrm{kHz}$, and illustrative efficiency 0.85, the stationary reference gives $G_p = 1.11065587$, $G_c = 0.11356221$ and a detected ratio $-0.7302943383\,\mathrm{dB}$ at $1\,\mathrm{MHz}$. These inputs specify a mathematical fixture, not measured density, collisions or detector calibration. An $8\,\mu\mathrm{W}$ periodic-seed calculation changes this ratio by about $10^{-4}\,\mathrm{dB}$, while deleting cross-sector correlations changes a $0.1\,\mathrm{MHz}$ example by $0.039\,\mathrm{dB}$. Individual noise components can differ by about 40% despite a small final ratio change. These observations do not give a global finite-seed error bound.

Gaussian photocounting adds a separate fourth-moment closure to the exact atomic second-moment QRT. Beyond the bright-seed linearization, the spontaneous mean and quadratic fluctuation intensity enter both PSD and SQL. A declared finite-band calculation at $8\,\mu\mathrm{W}$ changes the bright-seed result by $1.033 \times 10^{-5}\,\mathrm{dB}$, but increasing the optical collection bandwidth from 512 to 1024 MHz increases spontaneous collected flux by 5.016%. Thus no bound on an unfiltered optical tail follows from the small finite-band photocurrent correction.

26 Two moving phases and spatial optical modes

Define laboratory wavevectors $\mathbf{k}_0, \mathbf{k}_p, \mathbf{k}_c$ and $\mathbf{q} = \mathbf{k}_p - \mathbf{k}_0$, $\mathbf{Q} = \mathbf{k}_p + \mathbf{k}_c - 2\mathbf{k}_0$. Two phase coordinates are

$$\theta_1 = \nu t - \mathbf{q}\cdot\mathbf{r}, \qquad \theta_2 = -\mathbf{Q}\cdot\mathbf{r}, \qquad \dot{\boldsymbol{\theta}} = (\nu - \mathbf{q}\cdot\mathbf{v}, -\mathbf{Q}\cdot\mathbf{v}). \tag{85}$$

For couplings V_p, V_c, a consistent Hamiltonian is $h = h_0 + V_p e^{-\mathrm{i}\theta_1} + V_c e^{-\mathrm{i}(-\theta_1+\theta_2)} + \mathrm{h.c.}$. Expanding on the integer two-phase lattice yields

$$0 = \sum_{\mathbf{b}} \mathcal{L}_{\mathbf{a}-\mathbf{b}}\rho_{\mathbf{b}} + \mathrm{i}(\mathbf{a}\cdot\dot{\boldsymbol{\theta}})\rho_{\mathbf{a}}. \tag{86}$$

At exact $\mathbf{Q} = 0$, harmonics that represent the same physical phase must be coherently identified: combine $V_p + V_c^\dagger$ before solving. They are not independent photon ports. If $\mathbf{Q}\cdot\mathbf{v} = 0$ but $\mathbf{Q} \neq 0$, the second phase is a static spatial label and cannot simply be discarded.

The moving-grating term can be tested through the commutator identity. In a fixture with second-phase frequency $-1.123879 \times 10^6\,\mathrm{s}^{-1}$, freezing the moving grating omits the term $\dot{\theta}_2\partial_{\theta_2}K$ from the commutator identity. Diffusion remains PSD, but the normalized commutator defect is $2.96091344 \times 10^{-5}$. Increasing the lattice cutoff changes that residual by only 4.14×10^{-15}: the omission is physical, not a resolution error.

For spatial profiles $u_j(\mathbf{r}_\perp)$ normalized by $\int |u_j|^2 \mathrm{d}^2 r_\perp = 1$, the mode-projected response integrates the local polarization with both drive and collection profiles. Cross-mode noise comes from shared atoms and must retain those cross correlations. The limit $\mathbf{Q} \to 0$ is taken at fixed finite aperture, not by replacing its Fourier transform with an infinite-area delta function. A two-mode truncation does not establish convergence in the number of transverse modes.

Table 4: Conditional stationary spatial fixture: a $400 \times 300\,\mu$m rectangular aperture, prescribed pump, and the reciprocal model parameters of Sec. 25. Noise ratios refer to 1 MHz.

Angles (θ_p, θ_c) in mrad	G_p	G_c	S_- (dB)
$(0, 0)$	1.1106391761	0.1135452814	−0.7301956922
$(5, -5)$	1.0620337919	0.0650877602	−0.4328093597
$(5, -4)$	1.0277727399	0.0306514955	−0.2040641891

For the asymmetric-angle case, a transverse quadrature change $N_x = 4 \to 8$ moves the ratio by 3.5784×10^{-6} dB and fails a declared 10^{-6} dB tolerance; $8 \to 12$ changes it by 3.93×10^{-13} dB. Independent local QRT comparisons give relative noise and drift residuals 3.86×10^{-12} and 1.58×10^{-13}; the DC mean-field tangent test gives 5.77×10^{-10} and the global commutator test 1.38×10^{-11}. Only the RF points $0.1, 1, 4$ MHz are sampled. Neither these three points nor transverse quadrature convergence establishes a bandwidth or a full spatial mode basis.

27 Ballistic finite-age atoms and nonlocal covariance

An atom enters at $\mathbf{r}_{\mathrm{in}}$ with velocity $\mathbf{v}$ and prepared state ρ_{in}. Its age a labels $\mathbf{r}(a) = \mathbf{r}_{\mathrm{in}} + \mathbf{v}a$ and $\dot{\rho}(a) = \mathcal{L}(a)\rho(a)$ until its first exit at $a = \tau$. Let $\Phi(a, b)$ be the atomic fluctuation propagator. Then

$$\begin{aligned} C(a,b) &= \Phi(a,0)C_{\mathrm{in}}\Phi(b,0)^T \\ &\quad + \sum_r \int_0^{\min(a,b)} \mathrm{d}s\, \Phi(a,s)D^{(r)}(s)\Phi(b,s)^T. \end{aligned} \tag{87}$$

The first term is initial internal quantum noise; the second is noise from jumps during the flight. Both contain correlations between different ages. The field response is also retarded, containing $\Theta(a-b)C_{\mathrm{out}}(a)\Phi(a,b)B_{\mathrm{drive}}(b)$. A diagonal-in-age noise replacement loses shared-atom correlations even when each retained diagonal block is PSD.

One computes the integrated atomic output

$$Y_j = \int_0^\tau \mathrm{d}a\, e^{\mathrm{i}\omega_j a} O_j(a) \tag{88}$$

by augmenting the fluctuations with Y/τ. Its drift has blocks $\begin{pmatrix} A & 0 \\ e^{\mathrm{i}\omega a} C_{\mathrm{out}}/\tau & 0 \end{pmatrix}$ and its direct diffusion has D only in the atomic block. The integrated covariance is the final output block multiplied by τ^2; it includes all off-diagonal age correlations without storing an age-by-age kernel.

An independent adjoint formulation defines operators

$$\mathcal{A}_j(t) = \int_t^\tau \mathrm{d}s\, \mathcal{U}(s,t)^\dagger [O_j] e^{\mathrm{i}\omega_j s}, \quad \dot{\mathcal{A}}_j = -\mathcal{L}(t)^\dagger \mathcal{A}_j - e^{\mathrm{i}\omega_j t} O_j, \quad \mathcal{A}_j(\tau) = 0. \tag{89}$$

For $\mu_j = \mathrm{Tr}\, \rho_{\mathrm{in}} \mathcal{A}_j(0)$, its source-resolved greater covariance is

$$C^{>}_{\mathrm{in},jk} = \mathrm{Tr}[\rho_{\mathrm{in}} \mathcal{A}_j(0) \mathcal{A}_k(0)^\dagger] - \mu_j \mu_k^*, \tag{90}$$

$$C^{>}_{r,jk} = \int_0^\tau \mathrm{d}t\ \mathrm{Tr}\, \rho(t) [L_r^\dagger, \mathcal{A}_j(t)] [\mathcal{A}_k(t)^\dagger, L_r]. \tag{91}$$

The lesser covariance reverses the operator ordering. Linear response to an independent perturbation $V_k e^{-\mathrm{i}\nu_k t}$ follows from

$$\mathcal{R}_{jk} = -\mathrm{i} \int_0^\tau \mathrm{d}t\, e^{-\mathrm{i}\nu_k t}\, \mathrm{Tr}\{\rho(t) [\mathcal{A}_j(t), V_k]\}. \tag{92}$$

This reference constructs source covariances directly from commutators and the full density matrix, independently of the forward Einstein-matrix assembly. A separate two-time density-matrix QRT integral tests the complete integrated covariance. In a boundary-noise deletion counterexample all retained noise matrices stay PSD, but the commutator defect is 0.20196 and the output spectrum changes by 14.59%. Likewise, a diagonal-only kernel sum decreases toward zero under refinement whereas the full two-time trapezoidal integral converges to a finite covariance.

28 Thermal inflow, random arrivals and the unresolved cell closure

For independent Poisson arrivals with rate $\mathcal{J}$, a single-path integrated mark with connected covariance C and mean μ has spectrum

$$S_{\mathrm{stream}} = \mathcal{J}(C + \mu\mu^\dagger), \qquad \overline{N} = \mathcal{J}\tau. \tag{93}$$

The outer product is arrival-number noise, distinct from the internal atomic boundary covariance. For $O = I$ the internal noise vanishes, but random entry times still give $\mathcal{J} |\int_0^\tau e^{\mathrm{i}\omega a} \mathrm{d}a|^2$. With one random common entry phase, charges $(1, -1, -1, 1)$ retain only equal-charge cross terms. Average $\mu\mu^\dagger$ before subtracting or combining means; a zero phase-averaged μ does not imply zero number noise.

For a Maxwell velocity density f_M normalized in three dimensions and an inward boundary normal, the incoming flux measure is

$$\mathrm{d}\mathcal{J} = n f_M(\mathbf{v}) |\mathbf{v} \cdot \mathbf{n}| \mathrm{d}^2 r\, \mathrm{d}^3 v, \qquad \int \tau(\mathbf{r}, \mathbf{v}) \mathrm{d}\mathcal{J} = nV. \tag{94}$$

The first-exit lifetime τ depends on the boundary point and velocity. The occupancy identity checks geometric flux weights, not the oscillatory response/noise integral. Quadrature weights multiply path covariances once; squaring those weights or inserting density a second time is incorrect.

The declared thermal fixture is a square column of side $\sqrt{1.2 \times 10^{-7}} = 346.410\,\mu$m, length 12.5 mm, temperature 373 K and separately specified density $10^{18}\,\mathrm{m}^{-3}$. It injects an unpolarized internal state $\mathrm{diag}(5/12, 7/12, 0, 0)$ at the box boundary, with no upstream pumping history, re-entry or velocity-changing collisions. This is a mathematical boundary-value problem, not automatically

the physical boundary of a vapor cell. With a 530 μm pump waist, its boundary pump intensity is 0.652–0.808 of peak. A frozen zero-velocity atom at a side midpoint, exposed to the declared 0.6 W pump and 0.9 GHz detuning, changes its $F = 3$ population from 0.583333 to 0.645138, 0.869748, and 0.942976 after $0.1, 1, 5\,\mu$s. The corresponding Frobenius distances from the injected density matrix are $0.20293, 0.42950, 0.52642$. This is a boundary-sensitivity example, not an estimate of the experimental modeling error; it shows why entry-state preparation requires physical justification.

In a transported ensemble the optical equation contains a spatially nonlocal susceptibility kernel, and the output noise contains a double spatial integral:

$$\begin{aligned} \partial_z b(z) &= K_{\mathrm{geom}} b(z) + \int \mathrm{d}z'\, \mathcal{R}(z, z'; \Omega) b(z') + \eta(z), \\ W_{\mathrm{out}} &= \int \mathrm{d}z\, \mathrm{d}z'\, G(L, z) \mathcal{C}(z, z'; \Omega) G(L, z')^{\dagger}. \end{aligned} \tag{95}$$

An integrated finite-age response is not, by dimensional relabeling, a local matrix $M(\Omega)$. Nor may correlated trajectory segments be composed as independent local Gaussian channels. A solved, converged nonlocal optical boundary problem is required before a thermal transport spectrum becomes a cell-output squeezing prediction.

29 Exact constant-step convolution and controlled trajectory numerics

For a constant generator step, density evolution and source-resolved covariance can be integrated simultaneously. The scalar convolution

$$F(x, y; h) = \int_0^h \mathrm{d}u\, e^{x(h-u)} e^{yu} = \begin{cases} (e^{yh} - e^{xh})/(y - x), & y \neq x, \\ h e^{xh}, & y = x, \end{cases} \tag{96}$$

is evaluated with a stable divided-exponential expression near coincidence. If the density generator has eigenvalues λ_k, the augmented fluctuation drift has eigenvalues β_a, and each source $D_r(\rho)$ is linear in density, the particular covariance solution in that eigenbasis has entries

$$(\widetilde{C}_r(h))_{ab} = e^{(\beta_a + \beta_b^*)h} (\widetilde{C}_r(0))_{ab} + \sum_k \widetilde{D}_{r,abk}\, q_k F(\beta_a + \beta_b^*, \lambda_k; h). \tag{97}$$

The eigenoperators of the density generator can be complex and nonpositive. Hermitizing, clipping their eigenvalues, or normalizing their traces destroys the linear expansion. The implementation checks eigenbasis condition numbers (a declared guard 10^6), a roundoff amplification estimate $\epsilon \kappa(S)^2 \kappa(R) \leq 10^{-8}$, and eigen residuals $\leq 10^{-12}$; otherwise it uses a block exponential or another stable coordinate-free solve. Exact integration of each constant step does not make a changing pump envelope exact.

For an envelope $f(t)$ multiplying one Hamiltonian term, a fourth-order commutator-free composition uses nodes $c_{1,2} = 1/2 \mp \sqrt{3}/6$ and $a_{1,2} = (3 \mp 2\sqrt{3})/12$:

$$\begin{aligned} f_A = 2(a_2 f(c_1 h) + a_1 f(c_2 h)), \quad f_B = 2(a_1 f(c_1 h) + a_2 f(c_2 h)), \\ U_{\mathrm{CF4}}(h) = \exp[\tfrac{h}{2}(G_0 + f_B G_1)] \exp[\tfrac{h}{2}(G_0 + f_A G_1)]. \end{aligned} \tag{98}$$

Here the entire affine density/covariance lift follows the same composition. Although $a_1 < 0$, each duration is positive and each substep uses a Hermitian Hamiltonian with the same positive-rate jumps, so that each atomic substep is GKSL. Clipping a combined envelope changes the method and its order. General background on such propagation is given in Ref. [14].

A declared path gate requires three primary resolutions with two successive relative changes below 10^{-3}; agreement with a separately refined reference below 5×10^{-6}; and reference self-refinement below 2×10^{-6}. Gates apply to each RF matrix and each physical noise source, with a denominator

floor 128ϵ times a declared SI scale. That floor stabilizes a relative residual; it adds no physical noise. The tested $2\,\mu$s trajectory changes by 9.4185×10^{-4} from 4096 to 8192 CF4 steps, then by 1.6138×10^{-4} at 16384 steps, and differs from its independent reference by 8.6908×10^{-7}. For the separately sampled seed-811 path set, maximum successive changes are 5.347×10^{-6} and 2.135×10^{-7}, the forward/adjoint discrepancy is 6.08×10^{-8}, and reference refinement is 1.385×10^{-7}. These are path-solver results, not ensemble convergence.

29.1 A converged path is not a converged thermal ensemble

The reported thermal sample contains three independent scrambles, labelled 11, 211 and 811, of the same coarse $p = 2$ quadrature: 72 trajectories and 360 primary/reference solves in total. Directional relative comparisons use the second sample as denominator, so both directions are tested. The range across the six ordered comparisons is: The occupancy estimates divided by nV are 0.8302268543, 1.1102815979 and 0.8561468387. Path accuracy can be good while the ensemble is under-resolved; agreement of one integral cannot validate the others. There is no accepted refinement sequence for all response, ordered covariance, source-resolved covariance and number-noise matrices. Consequently neither the thermal covariance nor a derived SQL-normalized squeezing spectrum is certified by these samples.

Table 5: Thermal ensemble differences for the declared coarse inflow problem. Every ordered seed-pair comparison fails the 5% all-observable gate.

Quantity	Minimum difference (%)	Maximum difference (%)
Greater ordered covariance	12.8553	19.9038
Lesser ordered covariance	12.8928	19.9644
Greater covariance, by source	25.5642	43.6110
Lesser covariance, by source	22.1673	43.3337
Arrival-number contribution	5.6997	9.0633
Linear response	25.7841	34.3654

30 Readout precision and uncertainty conditional on the model

For detected currents I_p, I_c, complex electronic transfer functions H_p, H_c and subtraction weight w, the one-sided shot-noise PSD is

$$S^{(1)}_{\mathrm{SQL}}(f) = 2e\big(|H_p(f)|^2 I_p + w^2 |H_c(f)|^2 I_c\big). \tag{99}$$

The quantum photocurrent quadratic form uses the same mean currents, sideband ordering and electronic phases. Electronics noise and its cross-correlations are added with their measured signs. With $\langle x^2 \rangle = \int S_\Omega(\Omega)\mathrm{d}\Omega/(2\pi) = \int S_f(f)\mathrm{d}f$, the two-sided convention is $S_f(f) = S_\Omega(2\pi f)$: no extra 2π appears. For a real signal the positive-frequency one-sided PSD is twice that two-sided value. Numerator and SQL must use the same convention.

Average linear PSD and SQL through the actual RF resolution filter before forming their ratio, and convert to $10\log_{10}$ only afterward. Averaging dB values is a different observable. Finite bright-seed, Gaussian fourth-moment and optical collection-band assumptions are stated in Sec. 25; they are not consequences of the detector SQL identity.

For independently characterized parameters p with covariance Σ_p, linear propagation gives $\Sigma_y = J_p \Sigma_p J_p^T$, where $(J_p)_{ij} = \partial y_i / \partial p_j$. Joint sampling is preferable when responses are nonlinear, but must preserve parameter correlations, nonnegative rates and physically allowed density matrices. A parameter fixed by assumption is not independently measured merely because it was not fitted to the final squeezing trace. Uncertainty intervals conditioned on an unconverged thermal grid or unjustified boundary state do not cover those model errors. Measured gain, bandwidth, mode overlap, phase response and detector calibration must be held out from any fit used to claim predictive agreement.

Part VII
External detection model

31 External loss only after the cell

In the quadrature basis $\hat{\boldsymbol{R}} = (X_p, Y_p, X_c, Y_c)^T$, define

$$K_\eta = \mathrm{diag}(\sqrt{\eta_p}, \sqrt{\eta_p}, \sqrt{\eta_c}, \sqrt{\eta_c}), \qquad K_\ell = \mathrm{diag}(\sqrt{1-\eta_p}, \sqrt{1-\eta_p}, \sqrt{1-\eta_c}, \sqrt{1-\eta_c}).$$

Only loss physically located after the cell is applied:

$$V_{R,\mathrm{det}} = K_\eta V_{R,\mathrm{out}} K_\eta^T + K_\ell V_{\mathrm{vac}} K_\ell^T. \tag{100}$$

For symmetric efficiency,

$$V_{R,\mathrm{det}} = \eta_{\mathrm{ext}} V_{R,\mathrm{out}} + (1-\eta_{\mathrm{ext}}) V_{\mathrm{vac}}.$$

Atomic absorption and spontaneous emission remain distributed inside Eqs. (45)–(51). Cell optical-depth factors $\exp(-\mathrm{OD}_{p,c,\mathrm{seg}})$ are not applied a second time after propagation.

31.1 TMSV marginals, photon thinning, and what loss noise means

The following calculation is an analytic fixture for one collected spatiotemporal mode in each arm, with vacuum incident on an ideal two-mode squeezer. It is not a claim that the present dissipative GABES production transfer has this covariance. With an irrelevant squeezing phase suppressed, the two-mode squeezed vacuum (TMSV) is

$$|\psi_r\rangle = \sqrt{1-\lambda} \sum_{n=0}^{\infty} \lambda^{n/2} |n\rangle_p |n\rangle_c, \qquad \lambda = \tanh^2 r, \qquad \bar{n} = \frac{\lambda}{1-\lambda} = \sinh^2 r. \tag{101}$$

Tracing out either arm gives a geometric, equivalently single-mode thermal, number distribution [8],

$$\begin{aligned} \rho_p = \mathrm{Tr}_c |\psi_r\rangle\langle\psi_r| &= \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(1+\bar{n})^{n+1}} |n\rangle\langle n|, \\ \langle N_p \rangle = \bar{n}, &\qquad \mathrm{Var}\, N_p = \bar{n}(\bar{n}+1), \\ g_p^{(2)}(0) = \frac{\langle N_p(N_p-1)\rangle}{\bar{n}^2} &= 2. \end{aligned} \tag{102}$$

The same equations hold for c. Thus a pure bipartite state can have a mixed thermal marginal: the local number uncertainty is the uncertainty of the shared pair number, not an independently injected thermal field.

A quantum-limited post-cell attenuator is exactly a beam splitter with an independent vacuum environment mode [7, 6]:

$$\hat{a}_{j,\mathrm{det}} = \sqrt{\eta_j}\, \hat{a}_{j,\mathrm{out}} + \sqrt{1-\eta_j}\, \hat{v}_j, \qquad \langle \hat{v}_j^\dagger \hat{v}_j \rangle = 0, \qquad j \in \{p, c\}. \tag{103}$$

This is the annihilation-operator form of Eq. (100) with $V_{\mathrm{vac}} = I/2$. Conditional on $N = n$ photons before this beam splitter, the transmitted count obeys Bernoulli thinning,

$$K \mid N = n \sim \mathrm{Binomial}(n, \eta), \qquad \mathbb{E}[K \mid N] = \eta N, \qquad \mathrm{Var}(K \mid N) = \eta(1-\eta)N. \tag{104}$$

Applying the law of total variance to the thermal input gives

$$\begin{aligned} \mathbb{E}K &= \eta\bar{n}, \\ \mathrm{Var}\, K &= \underbrace{\eta^2\, \mathrm{Var}\, N}_{\text{attenuated source fluctuation}} + \underbrace{\eta(1-\eta)\mathbb{E}N}_{\text{partition noise}} = \eta\bar{n} + (\eta\bar{n})^2. \end{aligned} \tag{105}$$

Equivalently, the probability-generating function transforms as $G_K(z) = G_N(1-\eta+\eta z) = [1+\eta\bar{n}(1-z)]^{-1}$, so the output remains thermal with $\bar{n}_{\rm out} = \eta\bar{n}$. The second term in Eq. (105) is therefore binomial partition noise and is linear in flux (shot-noise-like); it does not mean that vacuum injects a Poisson-distributed photon population. The marginal remains thermal: vacuum loss scales every normally ordered factorial moment as

$$\left\langle (\hat{a}^\dagger_{\rm det})^m \hat{a}^m_{\rm det} \right\rangle = \eta^m \left\langle (\hat{a}^\dagger_{\rm out})^m \hat{a}^m_{\rm out} \right\rangle, \tag{106}$$

and hence leaves normalized $g^{(m)}$ invariant for nonzero transmitted mean; in particular $g^{(2)}(0) = 2$. At $\eta\bar{n} \ll 1$ the Fano factor $\mathrm{Var}\,K/\mathbb{E}K = 1 + \eta\bar{n}$ approaches one even though the normalized single-mode statistic remains thermal. This distinction explains why a strongly attenuated thermal beam can look shot-noise dominated without becoming a coherent state.

For independent loss ports on the two TMSV arms, the same pre-loss geometric pair number N is thinned separately. The resulting moments are

$$\langle K_j \rangle = \eta_j\bar{n}, \qquad \mathrm{Var}\,K_j = \eta_j\bar{n} + (\eta_j\bar{n})^2, \qquad \mathrm{Cov}(K_p, K_c) = \eta_p\eta_c\bar{n}(\bar{n}+1). \tag{107}$$

Consequently,

$$\mathrm{Var}(K_p - K_c) = \bar{n}[\eta_p(1-\eta_p) + \eta_c(1-\eta_c)] + (\eta_p - \eta_c)^2\bar{n}(\bar{n}+1), \tag{108}$$

$$\mathrm{Var}(K_p - wK_c) = \bar{n}\big[\eta_p(1-\eta_p) + w^2\eta_c(1-\eta_c)\big] + (\eta_p - w\eta_c)^2\bar{n}(\bar{n}+1). \tag{109}$$

For $\eta_p = \eta_c = \eta$, the common thermal pair-number fluctuation cancels from the raw difference. With $\mathrm{NRF} = \mathrm{Var}(K_p - K_c)/\langle K_p + K_c\rangle$, one obtains

$$\mathrm{Var}(K_p - K_c) = 2\eta(1-\eta)\bar{n}, \qquad \mathrm{NRF} = 1 - \eta. \tag{110}$$

Unequal efficiencies leak the common thermal fluctuation back into the raw difference through the last term of Eq. (108). For this ideal TMSV count model, $w = \eta_p/\eta_c$ cancels that term in Eq. (109); it is not a universal claim that this weight minimizes a detector-normalized spectrum. Once coherent carriers, electronic transfer, or a noncanonical source covariance is present, the optimal weight must be obtained from the full covariance and the matching SQL, as in the next section.

Each single-arm marginal of a coherently seeded ideal two-mode squeezer is a displaced thermal state,

$$\rho_j = D(\alpha_j)\rho_{\rm th}(\bar{n}_{\rm th})D^\dagger(\alpha_j), \qquad g^{(2)}_j(0) = 1 + \frac{\bar{n}^2_{\rm th} + 2\bar{n}_{\rm th}|\alpha_j|^2}{(\bar{n}_{\rm th} + |\alpha_j|^2)^2}. \tag{111}$$

It interpolates from 2 at zero displacement to 1 for a dominant coherent carrier. Thus hard-coding $g^{(2)} = 2$ is valid only for the declared vacuum-input, ideal single-mode TMSV fixture, not for the total bright seeded beam.

An attenuator whose unused port is thermal, with occupation n_E, is a thermal-loss channel rather than pure loss [6, 8]:

$$V_{j,\rm det} = \eta_j V_{j,\rm out} + (1-\eta_j)\left(n_E + \tfrac{1}{2}\right) I, \qquad \bar{n}_{j,\rm det} = \eta_j\bar{n}_{j,\rm out} + (1-\eta_j)n_E. \tag{112}$$

Here the environment adds thermal photons; ordinary optical path loss is normally modeled by $n_E \simeq 0$. Neither Eq. (103) nor Eq. (112) permits distributed in-cell absorption and spontaneous emission to be replaced, in general, by one beam splitter after an ideal source. Gain and attenuation are interleaved along z and need not commute. Their physical fluctuation prediction requires the distributed coupling $B(z, \Omega_{\rm SA})$ and symmetrized diffusion $D(z, \Omega_{\rm SA})$ in Eqs. (45)–(51); those microscopic inputs and measured collection conditions must be supplied by the chosen quantum model; the gain-only model does not supply them.

32 Electronic weighting and SQL normalization

The linearized photocurrent is

$$\delta\hat{i}_j(\Omega) \propto H_j(\Omega)\left[\alpha_j^*\delta\hat{a}_j(\Omega) + \alpha_j\delta\hat{a}_j^\dagger(-\Omega)\right], \qquad \delta\hat{i}_- = \delta\hat{i}_p - w(\Omega)\delta\hat{i}_c. \tag{113}$$

The normalized spectrum is

$$\boxed{S_-(\Omega) = \frac{\boldsymbol{m}^\dagger(\Omega)V_{R,\mathrm{det}}(\Omega)\boldsymbol{m}(\Omega) + S_{\mathrm{el}}(\Omega)}{S_{\mathrm{SQL}}^{(\mathrm{exp})}(\Omega)}.} \tag{114}$$

Here $\boldsymbol{m}(\Omega)$ is the four-component measurement vector obtained from Eq. (113); it contains the detected carrier amplitudes, H_p, H_c, their phases, and the factor $-w$ on the conjugate quadratures. Equations (51), (52), and (62) supply the in-cell block $V_{\mathrm{out}} = TV_{\mathrm{in}}T^\dagger + W$ and hence $V_{R,\mathrm{out}} = CV_\Xi C^\dagger$. Thus Eq. (114) is the single assembled observable: once the microscopic $M_q(\Omega_{\mathrm{SA}})$, $D(\Omega_{\mathrm{SA}})$, and measured detector functions are supplied, every remaining step is algebraic. With $V_{\mathrm{vac}} = I/2$ and independent detector vacua, the corresponding ideal electronic standard-quantum-limit (SQL) term is proportional to

$$S_{\mathrm{SQL}}(\Omega) \propto |H_p|^2\eta_p|\alpha_p|^2 + w^2|H_c|^2\eta_c|\alpha_c|^2,$$

before the experiment-specific electronic calibration. The SQL reference must use the same detected powers, electronic transfer functions, weight w, resolution bandwidth, and video bandwidth. The report must state whether electronic dark noise is included or subtracted.

Table 6: Loss and noise ledger.

Mechanism	Physical location	Correct treatment
Atomic absorption and spontaneous emission	distributed in the vapor	Complex response in M_q plus associated B and D
Cell windows, filters, and free propagation	after the cell	External beam-splitter efficiency η_{path}
Detector quantum efficiency	detector face	External beam-splitter efficiency η_{QE}
Electronic imbalance and bandwidth	after photodetection	$H_p(\Omega)$, $H_c(\Omega)$, $w(\Omega)$, and calibrated $S_{\mathrm{el}}(\Omega)$
Pump scattering into the detected mode	apparatus dependent	Measured technical-noise spectrum; not a universal atomic-loss formula

The ansatz

$$\mathcal{N}_{\mathrm{ps}} = \kappa(1 - \mathrm{e}^{-\mathrm{OD}_{\mathrm{pump}}})$$

is phenomenological. If used, it must be written $\mathcal{N}_{\mathrm{ps}}(\Omega) = \kappa(1 - \mathrm{e}^{-\mathrm{OD}_{\mathrm{pump}}})F_{\mathrm{ps}}(\Omega)$ and calibrated to the experimental spectral normalization. It is not used as a universal Langevin noise law in this report.

Part VIII
Validation

33 Validation hierarchy

Agreement between this algebra and the production program is an algebraic and numerical consistency check, not independent physical validation. The references independently check selected computational steps: the continued-fraction code assembles the diagnostic finite-Floquet blocks,

while the literal SI reference independently reconstructs Maxwell/Q propagation from a shared reduced susceptibility. It does not independently reproduce the atomic Hamiltonian or dissipator. The separate pump reference checks its trace-constrained null vector against an SVD construction, its weak response against an independent dense/direct solve, and its rotating frame against the finite-Floquet state, while still sharing the same four-level atomic assembly. The levels below are therefore reported separately.

34 Verification of the analytic identities

The closed forms in the response, propagation, and noise layers were checked against independent direct computations. Table 8 records the residuals. The resolvent test uses the engine′s own 16×16 Liouvillian at the literature operating point; the Lyapunov tests use randomized non-normal drifts together with deliberately constructed defective cases.

The two FAIL rows are labeled "basis" because they are not failures of the solution. The same quantity computed from Eq. (56), at the same drift, is accurate to 2×10^{-15}; only the eigenbasis resolution (58) degrades, and it degrades because S^{-1} does. As established in Sec. 18.2, φ_1 is entire; the singularity belongs to the eigenbasis representation. An implementation that diagonalizes M_q without a conditioning guard can return inaccurate results near a defective degeneracy, where the eigenvector matrix is ill conditioned.

35 Atomic pole spectrum at the operating point

Applying Eqs. (22)–(27) to the literature operating point ($\Delta/2\pi = +0.90\,\mathrm{GHz}$, $\delta/2\pi = -8\,\mathrm{MHz}$, $T = 121\,^{\circ}\mathrm{C}$, $\Omega_{\mathrm{pump}}/2\pi = 0.7076\,\mathrm{GHz}$) gives the pole structure in Table 9.

Power broadening widens the Raman pole fivefold. The slowest nonvanishing pole has half-width 0.5201 MHz, whereas the assumed ground-coherence rate is $\gamma_{gg}/2\pi = 0.1029\,\mathrm{MHz}$ from Eq. (12). The pole is broadened by a factor 5.06. Any estimate that identifies the Raman feature width with γ_{gg} is therefore low by a factor of 5.06 at this pump power. This illustrates the distinction in Sec. 10 that $-\mathrm{Re}\,\lambda_m$ is not a bare rate.

A narrow Raman feature remains in the one-dimensional axial average. The averaged response is not itself a finite pole sum. Rather, on the one-dimensional axial grid used for Table 9, the per-velocity Raman-pole centers remain at zero while their half-widths vary over 0.220–1.494 MHz: the two-photon combination cancels the common axial Doppler shift. This does *not* include the transverse term of Eq. (17), which supplies the 1.38 MHz rms center spread in the full two-dimensional geometry. The separate reference in Table 3 includes that term; the pole table itself remains one-dimensional. The Raman feature is therefore broadened but not smeared over the 247 MHz optical scale.

The separate scales do not determine a squeezing bandwidth. The per-velocity Raman poles supply Lorentzian half-widths of 0.22–1.49 MHz, while the non-collinear term supplies a Gaussian rms center spread of 1.38 MHz. These quantities have different definitions: a Lorentzian has no finite variance, and the maximum half-width of one velocity class is not the half-width of the averaged line. They therefore cannot be combined by a root-sum-of-squares rule. The measured 3.5 MHz bandwidth belongs to $S_{-}(\Omega_{\mathrm{SA}})$ after propagation and detection, not to a bare atomic pole. The two-dimensional mean-field response is evaluated by a separate reference, but a quantitative comparison still requires frequency-dependent $M_q(\Omega_{\mathrm{SA}})$ and $D(\Omega_{\mathrm{SA}})$ plus detector transfer functions. Until those are supplied, the pole table is a scale diagnosis and no numerical bandwidth gap is claimed.

36 Floquet-truncation convergence

Table 10 repeats the dressed-wavevector diagnostic calculation at the dressed-wavevector diagnostic optimum $\Delta/(2\pi) = -1.50\,\mathrm{GHz}$, $T = 110\,^{\circ}\mathrm{C}$ and approximately $\delta/(2\pi) = -280\,\mathrm{MHz}$. It deliberately retains the dispersion-double-counting dressed-wave-vector phase convention only to isolate Floquet convergence; because that convention double counts dispersion, these numbers are not the Option-A physical prediction. To prevent their diagnostic columns from being mistaken for canonical gains, the next two tables use $g_{c,\mathrm{diag}} \equiv |T_{21}|^2$ and $\Delta_{\mathrm{diag}} \equiv G_p - g_{c,\mathrm{diag}}$. Only their relative truncation behavior is used.

The relative changes from $N_F = 1$ to 2 are

$$\frac{|G_p^{(2)} - G_p^{(1)}|}{G_p^{(2)}} = 16.80\%, \qquad \frac{|g_{c,\mathrm{diag}}^{(2)} - g_{c,\mathrm{diag}}^{(1)}|}{g_{c,\mathrm{diag}}^{(2)}} = 12.97\%.$$

The diagnostic coefficient gap changes sign. Thus the $N_F = 1$ result is *qualitative and unconverged*. The changes from $N_F = 2$ to $N_F = 3$ are below $4\times10^{-7}\%$ at the tested point. Direct block solves and the independent assembly agree to 5.4×10^{-13} or better. Convergence at one operating point does not prove global convergence; this test must be repeated for every reported scan. The production path performs that full-scan $N_F/N_F - 1$ comparison and returns `UNCONVERGED` rather than silently promoting a failed scan.

For the propagation-only Option-A correction at the fixed literature point, the inherited approximate atomic response gives the results in Table 12. Here $N_F = 1 \to 2$ changes G_p by 0.4528% and G_c by 0.4906%; $N_F = 2$ and 3 agree at the displayed precision. This establishes Floquet convergence of that fixed finite-seed propagation benchmark. It does not use the pump-only reference; the comparison with that reference is reported separately below. The multidimensional-velocity approximation remains untouched. The displayed gains assume equal collected mode areas, but the conversion is explicit: $T_{\mathrm{can}} = Q^{-1}T_E Q$, $G_c = (\omega_c/\omega_p)|T_{\mathrm{can},21}|^2$, and $\mathcal{G}_{\mathrm{flux}} = G_p - |T_{\mathrm{can},21}|^2$.

The inherited weak-response implementation also retains a finite reference probe-seed. Its numerical linearity check is

Reference probe-seed power	G_p	G_c	$\mathcal{G}_{\mathrm{flux}}$
$2\,\mu$W	403.752756	406.102206	−2.342895
$4\,\mu$W	403.680319	406.019313	−2.332440
$8\,\mu$W	403.535497	405.853591	−2.311543

Between 2 and $8\,\mu$W, G_p changes by −0.05381%, G_c by −0.06122%, and the signed flux gap changes by −1.3382% relative to its $2\,\mu$W value. The absolute gains are weakly seed dependent, while the noncanonical flux gap is relatively sensitive. This supports numerical linearity with respect to the reference probe-seed at this fixed point only; by itself it does not validate the underlying pump-state approximation.

36.1 Pump-only reference and the finite-seed limit

The independent reference supplies an infinitesimal-seed comparison for the standard minus Raman branch; the production solver retains its finite seed. On the complete configured one-dimensional Doppler grid, four analysis frequencies $\Omega_{\mathrm{SA}}/2\pi = (0, 0.1, 1, 4)\,\mathrm{MHz}$ and five two-photon detunings $\delta/2\pi = (-30, -8, 0, 12, 40)\,\mathrm{MHz}$ give the diagnostics in Table 13.

The finite-reference susceptibility was then compared with the infinitesimal derivative after Doppler averaging all four complex response components over the same five detunings. Table 14 reports the worst component error normalized by the maximum pump-reference response.

The errors decrease monotonically, and a log–log fit to the last three rows has Rabi-amplitude slope 1.749. The finite range therefore supports convergence to the infinitesimal reference but does *not* establish an exact quadratic law. At the smallest amplitude, $N_F = 1$ gives a 4.485×10^{-3} worst

error, whereas $N_F = 2$ and 3 both give 9.138×10^{-6}; seed linearization and Floquet truncation must therefore be checked separately. The direct resolvent and diagonalizable pole/residue sum agree to 2.45×10^{-24} s or better over the declared analysis band, with eigenvector condition number 3.39. The projected zero-relative-frequency fixture has normalized residual 1.79×10^{-15}.

37 Physicality tests do not detect the Floquet truncation error

Since ρ_0 is the time average of a physical $\rho(t)$, it must be Hermitian and positive semidefinite, with unit trace. Table 15 checks those necessary conditions for the $N_F = 1$ truncation at both operating points, after the convergence failure above has already been quantified.

Every test passes, including at the dressed-wavevector diagnostic point where Table 10 shows a 16.8% error in G_p and a sign reversal of the diagnostic transfer-coefficient gap. State positivity is therefore insensitive to this truncation error: a valid density matrix can be quantitatively wrong. The exact-GKSL half-plane condition of Sec. 10.4 is an assembly invariant for the same reason, so neither check substitutes for explicit N_F convergence.

Table 7: Validation hierarchy. No combined error column is used.

Level	Test performed	Status
Algebraic consistency	Closed-form matrix exponential versus direct exponential; dimensional and frequency audits	PASS
Analytic-identity verification	Spectral resolvent expansion, Lyapunov closed form, block-exponential form, and the cubic characteristic polynomial against direct numerics	PASS (Table 8)
Numerical implementation consistency	Continued-fraction versus direct finite-block assembly for the dressed-wavevector diagnostic reference, plus production Maxwell/Q versus a literal SI propagation reference	PASS within shared atomic assumptions
Pump-only weak-response reference	Minus-branch static-frame/Floquet gauge parity, trace-one state, trace-zero finite-frequency and projected-DC response, finite-seed limit, and direct/pole-residue parity	PASS numerically within the four-level, one-dimensional-velocity model; separate slow reference, plus branch unsupported
Two-dimensional Raman-Doppler reference	Independent (v_z, v_x) Maxwell quadrature, lab-frequency invariance, analytic rms, collinear/equal-k limits, grid refinement, and cutoff refinement	PASS for the separate slow minus-branch reference; production remains one-dimensional
Truncation convergence	$N_F = 1, 2, 3$ direct/continued/kernel parity and scan-level adjacent-order checks	Production $N_F = 3/2$ gate implemented; diagnostic $N_F = 1$ FAIL
Microscopic reduced-model calculation	GKSL drift/diffusion, independent QRT, periodic sidebands and conditional spatial channels	Verified in declared finite models; full strong-drive Zeeman model pending
Thermal transport ensemble	Source-resolved forward/adjoint trajectory covariance and independent inflow scrambles	Selected paths converge; coarse ensemble fails (Table 5)
Experimental gain comparison	Reduced model evaluated at the documented ^{85}Rb operating point	performed; substantial discrepancy
Experimental bandwidth comparison	Full $S_-(\Omega_{\mathrm{SA}})$ versus the measured 3.5 MHz squeezing bandwidth	two-dimensional mean-field reference and conditional microscopic spectra available for declared models; converged thermal/nonlocal channel and measured detector response pending
Experimental squeezing-spectrum comparison	Full $S_-(\Omega_{\mathrm{SA}})$ using microscopic $D(\Omega_{\mathrm{SA}})$	not yet available

Table 8: Verification of the analytic results against direct numerics. Relative errors are Frobenius norms unless stated otherwise.

Identity	Reference computation	Relative error	Verdict
Spectral resolvent expansion, Eq. (22)	Direct linear solve on the 16×16 engine Liouvillian, $\Omega_{\mathrm{SA}}/2\pi = 0.1$–$4\,\mathrm{MHz}$	1.0×10^{-12}	PASS
Vanishing of the steady-state residue, Eq. (26)	$\langle l_0\|\mathcal{L}$ and $\langle l_0\|\mathcal{V}_k\|\rho_{\mathrm{ss}}\rangle$ on the engine Liouvillian	exactly zero in floating-point arithmetic	PASS
Coordinate-free solution, Eq. (56)	Simpson quadrature of Eq. (53); generic, defective, and coalescing drifts	2.2×10^{-15} (worst)	PASS
Eigenbasis resolution, Eq. (58)	Same quadrature, randomized non-normal M_q, PSD D	5.7×10^{-15}	PASS
Hermiticity and positivity of W	Same, invariants of Eq. (55)	1.0×10^{-16}; $\min \mathrm{eig}\, W > 0$	PASS
Eigenbasis resolution at a defective drift	Same quadrature, exact Jordan block	loses all accuracy	FAIL (basis)
Eigenbasis resolution near an exceptional point	Same quadrature, coalescing eigenvalues	1.7×10^{-1}	FAIL (basis)
Lyapunov superoperator spectrum, Eq. (57)	$\mathrm{spec}\,\mathcal{M}$ versus $\{\mu_m + \mu_n^*\}$, 200 random cases up to 4×4	1.5×10^{-13}	PASS
Block-exponential evaluation, Eq. (62)	Same quadrature, generic and defective drifts	8.5×10^{-15} (worst)	PASS
Cayley–Hamilton collapse, Eq. (38)	$\|N^2 - q^2 I\|$ and the resulting $\mathrm{e}^{M_{\mathrm{c1}} L}$ versus a direct exponential, including $q = 0$	1.5×10^{-14}	PASS
Characteristic polynomial, Eq. (123)	Direct eigenvalues of Eq. (122)	1.7×10^{-15}	PASS
Undamped limit, Eq. (124)	Roots versus $\Omega' = \sqrt{\Omega^2 + \Delta^2}$	exact to machine precision	PASS

Table 9: Slowest Liouvillian poles at the literature operating point, zero-velocity class. Half-width is $-\mathrm{Re}\,\lambda_m/2\pi$ and center is $-\mathrm{Im}\,\lambda_m/2\pi$. The Doppler column is the range of the half-width across the one-dimensional axial velocity grid.

Half-width [MHz]	Center [MHz]	Doppler range of half-width [MHz]	Assignment
0 (exact)	0	—	Steady state; null eigenvalue of Eq. (9)
0.5201	0	0.220–1.494	Ground-state Raman coherence, power broadened
0.5894	±190.05	0.311–1.494	Excited-state hyperfine coherence
3.3024	±1023.57	0.311–1.549	Optical/dressed coherence

Table 10: Floquet convergence of the dressed-wavevector diagnostic model at the fixed point $\delta/(2\pi) = -280\,\mathrm{MHz}$. Sectors $n = -N_F, \ldots, N_F$ are retained.

N_F	G_p	$g_{c,\mathrm{diag}}$	Δ_{diag}	$\arg \chi_{pc}$	$\arg \chi_{cp}$
1	22.5913	21.9687	+0.622609	178.142°	−179.333°
2	19.3415	19.4472	−0.105684	178.173°	−179.366°
3	19.3415	19.4472	−0.105684	178.173°	−179.366°

Table 11: Separate minimization of the gain-only objective. All observables in a row are evaluated at that row's δ^*. The scan spacing is 5 MHz.

N_F	$\delta^*/(2\pi)$ [MHz]	G_p	$g_{c,\mathrm{diag}}$	Δ_{diag}	$\arg\chi_{pc}$	$\arg\chi_{cp}$
1	−280	22.5913	21.9687	0.622609	178.142°	−179.333°
2	−275	8.97871	8.46840	0.510314	178.402°	−179.582°
3	−275	8.97871	8.46840	0.510314	178.402°	−179.582°

Table 12: Floquet convergence at the fixed literature point $\Delta/(2\pi) = +0.90$ GHz, $\delta/(2\pi) = -8$ MHz, and $T = 121\,^\circ$C. This is not an optimization; it is the production-like finite-seed path, not the separate pump-only reference.

N_F	G_p	G_c	$g_c^{(\Phi)}$	$\mathcal{G}_{\mathrm{flux}}$	$\arg\chi_{pc}$	$\arg\chi_{cp}$
1	401.70833	403.86243	403.85591	−2.147574	−179.886152°	179.814904°
2	403.53550	405.85359	405.84704	−2.311543	−179.885623°	179.814496°
3	403.53550	405.85359	405.84704	−2.311543	−179.885623°	179.814496°

Table 13: Executable pump-only state and trace-zero weak-response diagnostics.

Check	Maximum error or margin
Normalized pump-state equation residual	4.00×10^{-17}
Pump-state trace error	2.89×10^{-15}
Minimum pump-state eigenvalue	2.371×10^{-3}
Normalized weak-response equation residual, full 4×5 scan	7.32×10^{-17}
Weak-response trace error, full 4×5 scan	6.48×10^{-25}
Static-frame versus $N_F = 3$ pump-only Floquet state	8.37×10^{-15}
Projected zero-relative-frequency response residual	1.79×10^{-15}
Direct resolvent versus pole/residue sum, 0–4 MHz analysis band	2.45×10^{-24} s
Pole eigenvector condition number	3.39

Table 14: Finite-seed $N_F = 3$ approach to the pump-only infinitesimal reference.

Rabi-amplitude fraction	Seed power (μW)	Worst normalized error
1.00	8.0000	2.503×10^{-3}
0.30	0.7200	5.093×10^{-4}
0.10	0.0800	9.254×10^{-5}
0.03	0.0072	9.138×10^{-6}

Table 15: Physicality of the $N_F = 1$ truncated Floquet state. Minima are taken over the velocity grid and, for $\rho(t)$, over one beat period.

Quantity	Archived point	Literature point	Requirement
$\max \lVert\rho_0 - \rho_0^\dagger\rVert_\infty$	7.4×10^{-14}	3.3×10^{-14}	0
$\max \lvert \mathrm{Tr}\,\rho_0 - 1\rvert$	1.1×10^{-15}	5.6×10^{-15}	0
$\min \mathrm{eig}\,\rho_0$	$+7.90 \times 10^{-3}$	$+2.42 \times 10^{-3}$	≥ 0
$\min_t \mathrm{eig}\,\rho(t)$	$+5.63 \times 10^{-3}$	$+2.28 \times 10^{-3}$	≥ 0

38 Velocity-integration convergence

The following tests retain the same diagnostic propagation columns and use $N_F = 3$ with the one-dimensional Doppler integral. They validate only that quadrature, not the propagation convention or non-collinear geometry. At a fixed $5\sigma_v$ cutoff, velocity-step refinement gives the following errors:

Table 16: One-dimensional velocity-step convergence at fixed $|v_z| \leq 5\sigma_v$.

Δv [m/s]	Classes	G_p	$g_{c,\mathrm{diag}}$	Relative G_p error
10.00	195	18.936590	19.024328	$1.56 \times 10^{-3}\%$
5.00	389	18.936295	19.024021	$2.66 \times 10^{-7}\%$
2.50	777	18.936295	19.024021	$9.58 \times 10^{-8}\%$
1.25	1551	18.936295	19.024021	reference

Changing the velocity cutoff gives the following errors:

Table 17: One-dimensional cutoff convergence at $\Delta v = 2.5\,\mathrm{m/s}$.

Cutoff	G_p	$g_{c,\mathrm{diag}}$	Δ_{diag}	Relative G_p error
$3.0\sigma_v$	19.36195	19.46857	−0.10662	2.248%
$3.5\sigma_v$	19.11527	19.21041	−0.09514	0.945%
$4.0\sigma_v$	18.98214	19.07172	−0.08957	0.242%
$4.5\sigma_v$	18.93642	19.02414	−0.08773	$6.4 \times 10^{-4}\%$
$5.0\sigma_v$	18.93629	19.02402	−0.08773	reference

For the experimental phase-matching angle, the separate slow reference integrates Eq. (18) and passes the grid/cutoff criteria in Table 3. Its angular rms width $\sigma_{\delta,\theta}/(2\pi) = 1.38017\,\mathrm{MHz}$ is not negligible relative to $\gamma_{gg}/(2\pi) = 0.103\,\mathrm{MHz}$ and, as Table 9 shows, is comparable to the power-broadened Raman pole width itself. This result does not certify the one-dimensional finite-seed production scan.

39 Commutator-preservation test

At the literature operating point, assume for this diagnostic that the collected probe and conjugate have equal effective Gaussian mode areas, $A_p = A_c = 1.7105972 \times 10^{-7}\,\mathrm{m}^2$. Their distinct optical frequencies are retained in Q, rather than replaced by $Q \propto I$. The resulting photon-flux-normalized static drift in the symmetric mismatch gauge is

$$M_q = \begin{pmatrix} -1.795444 + 8.518638\mathrm{i} & -0.593347 + 297.229796\mathrm{i} \\ 0.962238 - 297.200894\mathrm{i} & 0.508740 + 60.465455\mathrm{i} \end{pmatrix}\ \mathrm{m}^{-1}. \tag{115}$$

If the actual conjugate collected-mode area differs from the equal-area audit assumption, $Q^{-1}M_EQ$ must be recomputed before applying the canonical metric J. The normalization operation itself is explicitly defined; the equal-area apparatus input remains conditional. For the numerical test only, factor the Hermitian matrix $K = -(M_qJ + JM_q^\dagger)$ as $K = BJ_FB^\dagger$. Here its eigenvalues are $0.285202\,\mathrm{m}^{-1}$ and $4.323166\,\mathrm{m}^{-1}$. Both are positive, so by Sec. 16 the displayed independent-vacuum algebraic completion with $J_F = I$ exists: at this point the drift requires loss-type reservoir channels. This construction proves that the drift can be embedded in a canonical channel; it is not a microscopic derivation of atomic noise.

Table 18: Canonical-commutator audit.

Test	Residual	Result
Option-A drift with reservoir omitted	$\max\lvert TJT^\dagger - J\rvert = 3.388$	FAIL
Local identity with the explicit Q transform	$\max\lvert M_qJ + JM_q^\dagger + BJ_FB^\dagger\rvert = 4.44 \times 10^{-16}\,\mathrm{m}^{-1}$	PASS conditional
Integrated identity through the cell, explicit Q	$\lVert TJT^\dagger + \int T_zBJ_FB^\dagger T_z^\dagger \mathrm{d}z - J\rVert_F / \lVert J\rVert_F = 3.71 \times 10^{-13}$	PASS conditional
Microscopic completion of this phenomenological static drift	Not established for this assumption set. The reciprocal reduced model has its own drift and jump-derived noise (Secs. 23–25)	Separate models

Because canonical commutator preservation constrains only the antisymmetric reservoir correlations, it fixes neither the symmetrized diffusion matrix $D(\Omega)$, nor its temperature, nor excess spontaneous-emission noise; the vacuum-fixture values below therefore serve as illustrative diagnostics for one chosen vacuum completion, not as bounds or Langevin-corrected physical predictions.

40 Gain-only diagnostic and its failure domain

The canonical ideal-amplifier limit must be separated from the phenomenological gain-only scan score. Define the conjugate photon-flux coefficient

$$g_c^{(\Phi)} \equiv |T_{\mathrm{cl},21}|^2 = \frac{\omega_p}{\omega_c} G_c. \tag{116}$$

For the two-mode map

$$\hat{a}_p^{\mathrm{out}} = \mu \hat{a}_p^{\mathrm{in}} + \nu \hat{a}_c^{\mathrm{in}\dagger}, \qquad |\mu|^2 = G_p, \quad |\nu|^2 = g_c^{(\Phi)},$$

the output commutator is $G_p - g_c^{(\Phi)} = \mathcal{G}_{\mathrm{flux}}$. Only when $\mathcal{G}_{\mathrm{flux}} = 1$ is this map canonical without additional reservoir operators. In that lossless limit the photon-number difference is conserved, and a bright coherent probe seed with a vacuum conjugate input gives

$$\boxed{S_{\mathrm{ideal}} = \frac{1}{G_p + g_c^{(\Phi)}} = \frac{1}{2G_p - 1}.} \tag{117}$$

Symmetric external efficiency η then gives

$$S(\eta) = 1 - \eta + \eta S_{\text{ideal}}. \tag{118}$$

For an ideal canonical amplifier with unequal efficiencies or a weighted difference, set $A = G_p$, $B = g_c^{(\Phi)}$, and $A - B = 1$. In units of the incident seed photon flux, the exact bright-seed variance and SQL are

$$\begin{aligned}\mathcal{V}_w &= \eta_p^2 A(A+B) + \eta_p(1-\eta_p)A \\ &\quad + w^2\left[\eta_c^2 B(A+B) + \eta_c(1-\eta_c)B\right] - 4w\eta_p\eta_c AB, \end{aligned} \tag{119}$$

$$S_w = \frac{\mathcal{V}_w}{\eta_p A + w^2\eta_c B}. \tag{120}$$

Here $w = 1$ is a raw difference and $w = \eta_p A/(\eta_c B)$ balances the detected DC levels. For unequal losses the weight that minimizes normalized noise is covariance dependent; it cannot in general be inferred from detected powers alone. Equation (120) reduces to Eq. (118) when $w = 1$ and $\eta_p = \eta_c$.

The gain-score optimizer instead evaluated the numerical score

$$S_{\text{gain}} = \text{clip}_{[0,1]} \frac{(G_p - g_c^{(\Phi)})^2}{G_p + g_c^{(\Phi)}}. \tag{121}$$

This score defines the gain-only model. It is not a covariance and its clipping does not prevent passive points from giving $S_{\text{gain}} < 1$; for example $(G_p, g_c^{(\Phi)}) = (0.2, 0.1)$ gives $S_{\text{gain}} = 0.033$. For the dissipative cell, $\mathcal{G}_{\text{flux}} \neq 1$ requires the reservoir channels of Secs. 16 and 18. Unequal detector weights must then be applied to the full covariance through Eq. (114), not reconstructed from the two mean gains.

41 Model point and experimental comparison

This benchmark applies the passive-limit-tested Option-A propagation signs and $N_F = 3$ to the inherited finite-reference-field four-level Floquet response. It is not an output of the pump-null-space and frequency-resolvent formulation of Parts II–III. It uses a $5\sigma_v$ one-dimensional velocity range and the documented point

$$\frac{\Delta}{2\pi} = +0.90\,\mathrm{GHz}, \quad \frac{\delta}{2\pi} = -8\,\mathrm{MHz}, \quad T = 121\,^\circ\mathrm{C}, \quad L = 12.5\,\mathrm{mm},$$

with 600 mW pump power, 8 μW probe-seed power, $w_{\mathrm{pump}} = 530\,\mu$m, $w_p = 330\,\mu$m, and $\theta = 0.32^\circ$. The radii are $1/\mathrm{e}^2$ intensity radii. The calculation omits the transverse two-photon Doppler average and therefore remains a reduced-model benchmark. In the same symmetric mismatch gauge used for Eq. (115), the transfer is

$$T_E = \begin{pmatrix} 18.839694 + 6.971473\mathrm{i} & -8.603522 + 18.218019\mathrm{i} \\ 8.625460 - 18.205907\mathrm{i} & 17.483771 + 10.224863\mathrm{i} \end{pmatrix}.$$

For equal collected mode areas, this gives

$$G_p = 403.53550, \quad G_c = 405.85359, \quad g_c^{(\Phi)} = 405.84704, \quad \mathcal{G}_{\mathrm{flux}} = -2.31154.$$

Here T_E is the electric-field transfer; applying the explicitly constructed $Q^{-1}T_E Q$ gives $g_c^{(\Phi)} = (\omega_p/\omega_c)G_c$ with $\omega_c/\omega_p = 1.00001614$. Five explicit model variants show the effect of inconsistent assumptions. The pair $(32.56066, 31.23207)$ uses the opposite susceptibility-propagation sign and fails passive Beer–Lambert attenuation. The pair $(5.61434, 4.66816)$ uses the passive sign but duplicates the manifold population factor and reverses the symmetric-frame mismatch sign. Removing only the population duplicate gives $(107.64682, 107.36704)$; its mismatch sign is still wrong. With those signs repaired but a coherence-only transit floor, the pair is $(1054.30596, 1061.88534)$. The phenomenological variant quoted above uses the trace-preserving thermal reload channel and gives $(403.53550, 405.85359)$. These are distinct assumption sets, not successive approximations with a common error bound. The residual $\ell_s = 0.74$ is held fixed. Even the last variant does not solve the manifold-normalization problem of Sec. 22. The first two inconsistent variants are retained as counterexamples, with explicit matrices and noise completions in Appendix J.

The static completion and detector model are flat by construction, so repeating them at several analysis frequencies adds no information. The measured features instead show what is missing: technical excess noise below a few hundred kHz, −8.7 dB from the IDS/SQL slope ratio at 150 kHz, a laser-source noise peak near 2 MHz, and loss of squeezing beyond roughly 3.5 MHz [1]. None can be reconstructed from two gains, so the table reports no physical $S_-(\Omega_{\mathrm{SA}})$ prediction.

The point $(\Delta/2\pi, \delta/2\pi, T) = (-1.50\,\mathrm{GHz}, -280\,\mathrm{MHz}, 110\,^\circ\mathrm{C})$ is classified only as a *numerical optimum under the dressed-wavevector diagnostic model*. It is not an experimentally established optimum. Likewise, the numerical proximity of the chosen-vacuum algebraic fixture to the measured −7.8 dB is not validation: the former contains no microscopic atomic diffusion or detector spectrum.

42 Required limiting cases

Table 19: Separation of mean-field gain and quantum-noise layers at the literature operating point, using explicit Q with an equal collected-area assumption. The symmetric external efficiency used in the diagnostic columns is $\eta_{\text{ext}} = 0.8694$.

Layer	G_p	G_c	S_-^{out}	S_-^{det}	Interpretation
Semiclassical reduced model	403.535	405.854	—	—	Mean-field gain only
Ideal Bogoliubov amplifier matched to G_p	403.535	402.542	0.001241 (−29.06 dB)	0.13168 (−8.80 dB)	Lossless limiting case, not the dissipative medium
Static independent-vacuum completion fixture	403.535	405.854	0.02414 (−16.17 dB)	0.15158 (−8.19 dB)	Canonical diagnostic with $w = 1$; D is not microscopic
Static independent-vacuum completion fixture, DC balanced	403.535	405.854	0.005233 (−22.81 dB)	—	Same diagnostic with $w \simeq 0.99429$; detector rematching not evaluated
Microscopic completion of this phenomenological assumption set	—	—	not yet available	not yet available	Not established; reciprocal model has different gains
Experiment [1]	111/8 = 13.875	109/8 = 13.625	—	0.166 (−7.8 dB)	Raw 8 μW input and 111/109 μW outputs; both power ratios are independently available

Table 20: Model–experiment comparison for the specified propagation signs, with duplicate manifold-population weighting removed.

Observable	Reduced model	Experiment or benchmark	Discrepancy
G_p	403.535	13.875 from raw powers	+2808.4%
G_c	405.854	13.625 from raw powers	+2878.7%
$\delta^*/(2\pi)$	−275 MHz in the dressed-wavevector diagnostic scan	−8 MHz measured operating point	−267 MHz; different objective and incomplete geometry make this nonpredictive
$\Delta^*/(2\pi)$	−1.50 GHz in the dressed-wavevector diagnostic scan	+0.90 GHz measured operating point	−2.40 GHz
Maximum squeezing	no physical prediction	−7.8 dB	No established microscopic apparatus completion of this phenomenological model; see Part VI
Squeezing bandwidth	no physical prediction; separate Lorentzian pole and Gaussian Doppler scales are tabulated	approximately 3.5 MHz	Two-dimensional mean-field reference complete; microscopic $S_-(\Omega_{\text{SA}})$ comparison pending (Sec. 35)

Table 21: Analytic limiting-case checks for the layered model.

Limit	Required result	Status
No atom–field coupling	$\chi_{ij} \to 0$ implies $G_p \to 1$, $G_c \to 0$, $D \to 0$, and $S_- \to 1$	PASS analytically
Ideal lossless parametric amplifier	$M_q J + J M_q^\dagger = 0$, $B = D = 0$, and $\mathcal{G}_{\mathrm{flux}} = 1$	PASS analytically
Bright probe-seeded ideal amplifier	$S_{\mathrm{ideal}} = 1/[G_p + g_c^{(\Phi)}]$, reducing to $1/(2G-1)$ when $\mathcal{G}_{\mathrm{flux}} = 1$	PASS analytically
Symmetric external efficiency	$S_-^{\mathrm{det}} = 1 - \eta + \eta S_-^{\mathrm{out}}$	PASS analytically
Zero phase-matching angle	$\theta \to 0$ removes $k\theta\sigma_v$ from the two-photon Doppler width	PASS analytically
Closed-field lossless limit	For a prescribed closed canonical field model, $B = D = 0$ and $TJT^\dagger = J$. Setting atomic jumps to zero alone does not remove initial atomic noise or nondecaying modes	PASS as a formal limit
Zero cell length	$W(0) = 0$ from Eq. (55), so $V_{\mathrm{out}} = V_{\mathrm{in}}$	PASS analytically and numerically
Undamped two-level drive	Poles at $0, \pm\mathrm{i}\sqrt{\Omega^2 + \Delta^2}$ from Eq. (123)	PASS numerically
Vanishing perturbation coupling operator	$\mathcal{V}_k \to 0$ makes every residue $A_m^{(jk)}$ vanish, so $\bar{\chi} \to 0$; taking only the probe amplitude to zero instead leaves a finite linear susceptibility	PASS analytically

43 Conclusions and validation status

The stationary resolvent, pole expansion with its defective-generator qualification, entire Lyapunov solution and post-cell detection map supply the analytic core. The microscopic extension derives ordered diffusion from GKSL jumps and checks it independently by QRT; it retains periodic cross-harmonic correlations, the physical sideband space, transverse phases, finite-age boundary noise and Poisson arrival noise. These are explicit physical constraints on reduced models, not fitted squeezing formulas.

Admissibility and numerical accuracy remain separate. Positive density matrices do not certify Floquet convergence; PSD noise does not certify commutators or channel CP; accurate individual trajectories do not certify a thermal integral. The coarse thermal samples in Table 5 fail their declared agreement gate. Their injected internal state is an additional boundary assumption, and a nonlocal Maxwell solution is required to turn transported atomic kernels into cell-output fields.

The phenomenological gain model substantially disagrees with measured output powers at the documented experimental setting. Conditional quantum results from a reciprocal four-state model use a different normalization and declared inputs and do not remove that discrepancy by themselves. A predictive apparatus spectrum requires a justified Zeeman/polarization reduction, independently characterized density and reservoirs, physically prepared inflow, converged thermal and spatial integrals, a solved optical boundary problem, and measured collection and detector functions. The pole, Doppler and conditional noise examples in this document constrain that work; none establishes the experimental squeezing bandwidth or noise level.

A Analytically solvable limit: the damped two-level rotor

The full four-level Liouvillian is 16×16 and its characteristic polynomial has degree 16, so its roots are computed numerically. A reduced two-level case supplies the physical labels used in Sec. 10. With longitudinal and transverse relaxation times T_1 and T_2, detuning Δ, and Rabi frequency Ω, the Bloch generator in the (u, v, w) basis is

$$M_{\mathrm{B}} = \begin{pmatrix} -1/T_2 & -\Delta & 0 \\ \Delta & -1/T_2 & \Omega \\ 0 & -\Omega & -1/T_1 \end{pmatrix}, \tag{122}$$

with the factored characteristic polynomial

$$\boxed{\left(\lambda + \tfrac{1}{T_2}\right)\left[\left(\lambda + \tfrac{1}{T_2}\right)\left(\lambda + \tfrac{1}{T_1}\right) + \Omega^2\right] + \Delta^2\left(\lambda + \tfrac{1}{T_1}\right) = 0.} \tag{123}$$

The limiting cases are:

- **Undriven**, $\Omega = 0$: the population pole is $-1/T_1$ and the free-precession pair is $-1/T_2 \pm \mathrm{i}\Delta$.
- **Resonant**, $\Delta = 0$: the u component has pole $-1/T_2$; the remaining quadratic gives a damped-Rabi pair only when $|\Omega| > |T_1^{-1} - T_2^{-1}|/2$, and otherwise gives two additional real overdamped branches.
- **Undamped**, $T_1, T_2 \to \infty$:

$$\lambda_0 = 0, \qquad \lambda_\pm = \pm \mathrm{i}\Omega', \qquad \Omega' = \sqrt{\Omega^2 + \Delta^2}. \tag{124}$$

Thus the roots comprise one population-recovery branch and either a complex-conjugate damped-Rabi pair or two additional real overdamped branches. This is the structure named by Eq. (27) in the multilevel case. Equation (123) follows Torrey's classical transient-nutation analysis [9].

Table 22: Dimensional audit in SI units.

Quantity	Dimension	Check
$\Delta, \delta, \Gamma, \Omega_{\mathrm{SA}}, \Delta_e, \lambda_m$	s^{-1}	Angular frequencies and Liouvillian poles
$\bar{\chi}_{ij}$; $Nd^2/(\epsilon_0\hbar)$	s; s^{-1}	Their product is dimensionless χ_{ij}
$M_{\mathrm{cl}}, M_q, D, K$	m^{-1}	Spatial drift, diffusion, and commutator defect per unit length
$\mu_\pm$	m^{-1}	Eigenvalues of a spatial generator
$W(L)$	1	$\int D\,\mathrm{d}z$ with dimensionless covariance
$V(x;\sigma,\gamma)$	s	$-\pi V$ is an inverse angular frequency
G_p, G_c, S_-	1	Power gains and normalized noise spectrum

B Dimensional-consistency audit

C Assumptions and validity domains

Table 23: Assumption and validity-domain ledger.

Layer	Assumption	Validity domain and failure mode
Atomic basis	Four hyperfine manifolds with Zeeman structure averaged into effective strengths	Hyperfine-scale qualitative trends; fails for polarization, Zeeman-resolved optical pumping, Zeeman dark states, and magnetic response
Pump-only weak-response reference	Undepleted prescribed pump, static four-level Markovian Liouvillian, trace-one null space, and trace-zero Nambu resolvent	Executable slow reference for the minus Raman branch only; not used by the current gain tables, plus branch unsupported, and controlled only under the stated phenomenological decay channels; a separate 2-D wrapper is available
Current gain-state path	Trace-one finite-seed density matrix; production $N_F \geq 3$ with an $N_F - 1$ full-scan gate, audit $N_F = 1, 2, 3$	Not a pump-only state; convergence and finite-seed back-action must be reported separately
Rate/Sylvester model	Populations solved without coherence feedback	Weak saturation or rapid dephasing; not an exact strong-pump steady state
Weak fields	Probe and conjugate enter to first order	Valid below the onset of pump depletion, gain saturation, and nonlinear probe back-action
Pole expansion	$\mathcal{L}$ is diagonalizable	Exact where it holds; defective Liouvillian eigenvalues require a Jordan or contour form with higher-order poles, and eigen-residues become ill conditioned near defective or nearly defective points

Continued on next page

Layer	Assumption	Validity domain and failure mode
Lorentzian reading of poles	Residues vary slowly over the analysis band	Overlapping poles with comparable residues interfere and do not decompose into independent visible peaks
Floquet expansion	Finite N_F	Must be increased until observables converge; $N_F = 1$ fails at the dressed-wavevector diagnostic point, and no positivity test detects the failure
Velocity average	Maxwell distribution, independent Cartesian components	The separate minus-branch reference resolves (v_z, v_x) for one declared angle; the finite-seed production average remains one-dimensional and omits beam divergence
Classical propagation	z-independent coefficients and fixed density, temperature, and pump	Closed-form exponential; fails with longitudinal pump attenuation, density gradients, or depletion
Lyapunov solution	Constant M_q, D over the integration interval	No condition on M_q; use the segmented recursion for varying coefficients. Only the eigenbasis *resolution* additionally needs a well-conditioned S
Phase convention	Diagonal complex susceptibility plus vacuum/geometric mismatch only	Avoids double counting of $\mathrm{Re}\,\chi$
Quantum propagation	Linear Gaussian Langevin channel	Requires a microscopic reservoir matrix and appropriate Markov approximation
Static vacuum completion	Minimal-port independent-vacuum factorization of the drift	Commutator fixture only; neither a universal Caves-bound claim nor an atomic-noise prediction, and frequency independent by construction
Detection	External path and detector losses act after the cell	Valid only for losses physically distinct from distributed atomic absorption
Electronics	Linear transfer functions and calibrated SQL reference	Requires measured $H_p, H_c, w, S_{\mathrm{el}}$ at every analysis frequency

D Parameter provenance and sensitivity

The classifications below distinguish quantities measured for the cited apparatus, assumed in the reduced model, fitted or inherited phenomenologically, and algebraically derived. A dash means that the parameter does not enter that model layer. Unless an experimental uncertainty and its source are stated explicitly, a numerical ″audit range″ below is only an illustrative sensitivity sweep; it is not a confidence interval, prior, or measurement uncertainty.

Table 24: Measured, assumed, fitted, inherited, and derived parameters.

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_p, G_c, S_-
$L = 12.5\,\mathrm{mm}$	Vapor interaction length	Measured; experimental cell [1]	Use manufacturing/ calibration uncertainty when available	Gains depend through $\exp(M_{\mathrm{cl}}L)$; noise through $W(L)$ of Eq. (56)
$T = 121\,^{\circ}\mathrm{C}$	Cell-reservoir operating temperature	Experimentally measured operating point [1]	Not a fitted model optimum	Strong density and Doppler dependence; must be rescanned
$P_{\mathrm{pump}} = 600\,\mathrm{mW}$, $P_0 = 8\,\mu\mathrm{W}$	Incident pump and probe-seed powers used for squeezing	Experimentally measured operating point [1]	The probe-seed was approximately $10\,\mu$W in the apparatus description; $8\,\mu$W for the cited squeezing trace, with raw outputs $P_p = 111\,\mu$W and $P_c = 109\,\mu$W	Pump changes the steady state and power broadens the poles (Table 9); the probe-seed sets the bright-beam linearization and the SQL
$w_{\mathrm{pump}} = 530\,\mu\mathrm{m}$, $w_p = 330\,\mu\mathrm{m}$	$1/\mathrm{e}^2$ intensity radii, not diameters	Measured [1]	Use the reported beam convention	Set Rabi frequency and spatial overlap
$\theta = 0.32^{\circ}$	Phase-matching angle	Measured geometry [1]	Zero-angle check required	Produces 1.38 MHz rms angular Raman broadening at 121 °C, comparable to the power-broadened Raman pole width
$\Delta/(2\pi) = +0.90\,\mathrm{GHz}$, $\delta/(2\pi) = -8\,\mathrm{MHz}$	Experimental optical and two-photon operating detunings	Experimentally measured operating point [1]	Not identified with the diagnostic numerical optimum	Corrected-sign, one-use-normalized reduced model gives $G_p = 403.535$, $G_c = 405.854$ after the thermal transit reset; the 0.74 residual is not refitted
$\Gamma/(2\pi) = 5.746\,\mathrm{MHz}$	Natural optical linewidth used in the model	Literature benchmark	Ordinary-frequency value converted once to angular rate	Sets optical resolvents and Voigt width
$\gamma_{\mathrm{opt}}/(2\pi) = 0.742\,\mathrm{MHz}$ at 121 °C	Added collisional optical half-width $\frac{1}{2}\beta N$	Derived from the density and the self-broadening coefficient	Scales linearly with $N(T)$	Broadens the optical poles of Table 9
$\sigma_v = 193.7\,\mathrm{m/s}$ at 110 °C; 196.5 m/s at 121 °C	One-axis thermal rms speed	Derived from Maxwell statistics	$\sigma_v = \sqrt{k_B T/m}$	Sets both one-photon and mismatch-projected Doppler widths (one-photon rms 243.6 MHz and 247.2 MHz; angular-mismatch rms 1.36 MHz and 1.38 MHz, at 110 °C and 121 °C, respectively)

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Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_p, G_c, S_-
$\ell_s = 0.74$	Effective line-strength or overlap multiplier	Inherited from code; phenomenological coefficient with unknown fit provenance	Original fitting record unavailable; illustrative ±10% sweep 0.666–0.814, not an uncertainty interval	At fixed $N_F = 3$ atomic response, (G_p, G_c) changes from $(188.447, 189.054)$ through $(403.535, 405.854)$ to $(857.836, 863.691)$; physical S_- remains unavailable
$\kappa = 0.1$	Coupling of absorbed pump fraction to detected pump-scattering noise	Inherited from code; assumed phenomenological coefficient	No documented fit; illustrative sweep 0–0.2, not an uncertainty interval, pending apparatus calibration	No effect on G_p, G_c; the diagnostic obeys $N_{\mathrm{ps}} = \kappa(1 - \exp[-0.149609])$ with slope 0.138956, but normalized $S_-(\Omega)$ requires F_{ps} and detector calibration
$\gamma_t/(2\pi) =$ 100 kHz at 121 °C ($\gamma_{gg,\mathrm{total}}/(2\pi) =$ 102.882 kHz)	Thermal transit-reset input; the total ground-coherence rate also contains a 2.882 kHz collisional pure-dephasing term	Inherited from code; assumed phenomenological rate	Illustrative $\gamma_t/(2\pi)$ sweep 90–110 kHz, not an uncertainty interval; independent linewidth fit absent	Recomputed $N_F = 3$ endpoints give $(G_p, G_c) =$ $(440.250, 442.733)$ at $\gamma_t/(2\pi) = 90$ kHz, $(403.535, 405.854)$ at 100 kHz, and $(370.626, 372.790)$ at 110 kHz (total rates 92.882, 102.882, and 112.882 kHz, respectively); the pole width is not a direct measurement of this parameter
$1 - \eta_{\mathrm{path}} =$ 5.5(2)%	Post-cell optical-path loss	Measured [1]	Allowed range 5.3–5.7%; not cell absorption	No effect on gains; for symmetric loss $\partial S_{\mathrm{det}}/\partial\eta = S_{\mathrm{out}} - 1$
$1 - \eta_{\mathrm{det}} = 8.0(2)\%$	Detector and remaining system inefficiency	Measured [1]	Allowed range 7.8–8.2%	No effect on gains; combines multiplicatively with path efficiency
$\eta_{\mathrm{ext}} = 0.8694$	Illustrative total external efficiency 0.945×0.920	Derived from measured losses	The paper also quotes total loss 13.5(2)%; definitions must be reconciled	Maps S_{out} to $1 - \eta_{\mathrm{ext}} + \eta_{\mathrm{ext}} S_{\mathrm{out}}$
N_F	Retained Floquet sideband order	Numerical truncation	$N_F = 3$ production default with a full-scan $N_F = 2$ comparison	$N_F = 1$ changes the diagnostic coefficients by 13–17% and reverses the sign of their gap, without violating any positivity test
$\ell_{\mathrm{eff}} = 0.74/12$	Residual line/overlap factor times the sublevel average	Population supplied by trace-normalized ρ_{ss}; duplicate removed	$\ell_{\mathrm{eff}} = 0.0616667$; see Appendix E	Algebraic factor count only; separate manifold test in Sec. 22 rejects a universal common factor

No entry involving ℓ_s, κ, or the adopted γ_{gg} is described as exact. Their source records and any apparatus-level fitting provenance remain required metadata. The atomic solve also begins from an already-declared cell-input probe power; it contains no EOM V_π, insertion-loss, sideband-fraction, or impurity calibration. Missing EOM provenance therefore blocks an apparatus-level mapping from RF settings to that optical input and blocks any EOM-noise claim.

These missing records activate the claim gate rather than being filled by an illustrative sweep: algebraic identities and finite-model convergence may still be reported, but quantitative apparatus-level gain remains UNSUPPORTED and physical $S_-(\Omega_{\mathrm{SA}})$ remains UNAVAILABLE. A calibration-point agreement, if later obtained, may not promote either claim without a predeclared held-out comparison.

E Dipole and population normalization ledger

Let $d_{\mathrm{D1}} = \langle J_e \| er \| J_g \rangle$ be the fine-structure reduced matrix element. The reduced coefficient is not the polarization-summed hyperfine strength. It is the following *AutoOD $q = 0$ convention*:

$$\mathcal{S}_0(F_g, F_e) = \sum_{m=-\min(F_g,F_e)}^{+\min(F_g,F_e)} \begin{pmatrix} F_e & 1 & F_g \\ m & 0 & -m \end{pmatrix}^2, \tag{125}$$

$$C_F^2 = (2F_g+1)(2F_e+1)(2J_g+1)(2J_e+1)(2L_g+1)\mathcal{S}_0(F_g,F_e) \times \begin{Bmatrix} J_g & J_e & 1 \\ F_e & F_g & I \end{Bmatrix}^2 \begin{Bmatrix} L_g & L_e & 1 \\ J_e & J_g & S \end{Bmatrix}^2. \tag{126}$$

For $I = 5/2$, $S = 1/2$, $L_g = 0$, $L_e = 1$, and $J_g = J_e = 1/2$, the result is $\mathcal{S}_0 = 1/3$, which gives the stored exact values

$$C_F^2 = \{10, 35, 35, 28\}/81$$

for $(F_g, F_e) = (2,2), (2,3), (3,2), (3,3)$, respectively.

The AutoOD orbital reduced dipole and the fine-structure reduced dipole used by GABES obey $|d_{\mathrm{orb}}|^2 = 3|d_{\mathrm{D1}}|^2$ in this D1 convention. Therefore

$$|d_{\mathrm{eff}}|^2 = C_F^2 |d_{\mathrm{orb}}|^2 = 3C_F^2 |d_{\mathrm{D1}}|^2,$$

and the dimensionless amplitude multiplier that accompanies d_{D1} is

$$a_{ge} = \sqrt{3C_F^2}. \tag{127}$$

This derivation explains the code factor but also its limitation: Eq. (126) is a $q = 0$, Zeeman-averaged absorption convention, not a universal incoherent sum over $q = 0, \pm 1$. A polarization-resolved Liouvillian must replace it with explicit Clebsch–Gordan amplitudes and must not multiply by a_{ge} again.

A double-population diagnostic convention assembles the factor as

$$\ell_{\mathrm{eff}}^{\mathrm{old}} = \ell_s \frac{p_{F=3}}{2(2I+1)} = 0.74 \times \frac{7}{144} = 0.0359722. \tag{128}$$

This double-counted the manifold population: the trace-normalized ρ_{ss} already supplies the actual, pump-modified population of the representative $F = 2, 3$ states. Removing that explicit duplicate gives the phenomenological convention

$$\boxed{\ell_{\mathrm{eff}} = \ell_s \frac{1}{2(2I+1)} = \frac{0.74}{12} = 0.0616667.} \tag{129}$$

Removing an explicit duplicate is necessary but does not establish the correct reduction of each populated hyperfine manifold. Equation (75) requires $3C_F^2 p_F/(2F+1)$, not $3C_F^2 p_F/12$. Only after

setting $p_F = (2F+1)/12$ does the exact thermal weak-absorption factor reduce to $3C_F^2/12$. Thus Eq. (129) is retained as the definition of the phenomenological gain fixture, not a validated uniform factor for independently pumped $F = 2, 3$ populations. The residual ℓ_s remains assumed and cannot supply the missing manifold or polarization physics by a scalar fit.

The normalization rule for future implementations is:

1. use the fine-structure reduced dipole matrix element once;
2. use either explicit Clebsch–Gordan matrix elements or the averaged a_{ge}, never both;
3. include isotope abundance once in N;
4. obtain ground-manifold population either from ρ_{ss} or an external p_F, never both;
5. include spontaneous-emission branching ratios in the collapse operators, not again in susceptibility amplitudes; and
6. apply external optical efficiencies only in the detection layer.

Table 25: Code-path normalization audit for the current reduced FWM solver.

Factor	Where it enters	Verdict
Reduced D1 dipole d_{D1}	Once in the macroscopic conversion in `gabes/observables.py`	no duplicate found
Hyperfine/Zeeman strength	$\sqrt{3C_F^2}$ in the weak-field drive and once in polarization readout; their product gives $3C_F^2$. It is not also present in `physical_coupling_norm`.	no algebraic duplicate found
Isotope abundance	The FWM path uses the pure-^{85}Rb density from `hyperfine.number_density`; it does not multiply the natural 0.7217 abundance.	no duplicate in this path
Ground-manifold population	ρ_{ss} is trace normalized over the two lumped ground manifolds and supplies the population once. `physical_coupling_norm` supplies only $1/[2(2I+1)]$.	duplicate removed; common manifold factor not validated
Spontaneous branching	The C_F^2-normalized decay fractions occur in Liouvillian collapse channels; no additional branching multiplier is applied in `chi_phys`.	no duplicate found
Residual ℓ_s	Multiplies the macroscopic coupling once.	inherited, not derived

The coefficient ledger distinguishes algebraic duplicate removal from a physical manifold reduction. A representative-state coefficient identity is regression-tested, but it does not validate the factor for independently pumped manifold populations. Dynamical tests inspect the pump-state ρ_{ss} and its trace-zero weak response. Those dynamical tests do not supply an independent absolute-OD scale and therefore are not, by themselves, a full weak-probe OD validation. Any future end-to-end OD comparison must count each population factor exactly once.

F Four-level model limitations and unresolved physics

The four-hyperfine-level model is suitable for qualitative gain structure, reduced hyperfine-scale intuition, and comparison with the existing reduced simulation. It is not automatically sufficient for absolute squeezing, polarization-sensitive effects, magnetic-field response, quantitative spontaneous-emission noise, or accurate Raman linewidths.

The following physics is absent from, or not jointly established in, a converged apparatus model. Some items are included conditionally in the reduced constructions of Part VI:

- polarization-dependent Clebsch–Gordan pathways and explicit $\sigma^{\pm}$ and π selection rules;

- Zeeman optical pumping, ground-state orientation and alignment, dark-state formation, and magnetic-field-dependent coherences;
- interference among distinct m_F pathways and polarization-dependent spontaneous emission;
- spatial pump depletion, longitudinal attenuation, transverse Gaussian averaging, diffraction, mode mismatch, and cell-window etalons;
- buffer-gas or wall-collision dynamics, transit-time distributions, radiation trapping, and non-Markovian reservoirs;
- multidimensional velocity-changing collisions, beam angular distributions, and promotion of the separate non-collinear Raman-Doppler reference into the finite-seed segmented production path;
- a converged apparatus-level microscopic diffusion and collected fluorescence model; reduced-model jump diffusion is derived in Part VI;
- detector transfer functions, electronic cross-talk, common-mode rejection, and the apparatus-specific 2 MHz laser-source noise peak; and
- an independently measured SQL calibration and gain definition matched exactly to the theoretical modes.

None of these can be assumed small. The gain discrepancy of Table 20 and the 1.36–1.38 MHz angular Doppler scale, which is comparable to the power-broadened Raman pole width itself, are order-unity effects; the present calculation does not yet define a squeezing-bandwidth prediction.

G Implementation notes for the analytic layers

The following are the practical requirements implied by Secs. 10 and 18.

1. **Do not diagonalize M_q without a guard.** Use Eq. (62) by default. If Eq. (58) is used for interpretation, check cond(S) and fall back when it exceeds a threshold; the two forms agree to 10^{-15} in the well-conditioned regime.
2. **Assert the invariants.** W must be Hermitian and positive semidefinite for any Hermitian, positive-semidefinite D, and $W(0) = 0$. These are cheap tests that catch sign and conjugation errors in the creation/annihilation bookkeeping.
3. **Treat the stationary subspace explicitly at zero frequency.** For $\Omega \neq 0$ the unique trace-mode residue vanishes by Eq. (26). At $\Omega = 0$, solve on the trace-zero subspace or use the Drazin inverse; do not evaluate a formal 0/0 term. If more than one stationary mode exists, project the full stationary subspace and check every zero-mode residue.
4. **Track residues, not only eigenvalues.** A narrow pole with a negligible residue is not a visible spectral feature. Ranking poles by $|A_m|/\Gamma_m$ rather than by Γ_m alone identifies which ones actually shape $S_-(\Omega_{\mathrm{SA}})$.
5. **Recompute K after any normalization change.** Equation (50) is contracted with J and is therefore basis dependent through Eq. (7).
6. **Report the analysis frequency with every spectrum.** The Ω_{SA} dependence enters both M_q and D; a spectrum quoted without its frequency grid cannot be compared with a measurement.

H Reproducibility

The diagnostic Floquet and velocity-grid tables and the separate Option-A literature-point transfer, Maxwell/Q parity, commutator fixture, and model-layer comparison, plus the ℓ_s, κ, transit-floor, and two-dimensional Raman-Doppler grid/cutoff sensitivity blocks, are generated by `analysis/`

`squeezing/analytic_reconstruction/convergence_audit.py`. Its human-readable and machine-readable outputs are `generated/convergence_audit.md` and `generated/convergence_audit.json`. Those files retain the separate dressed-k diagnostic scans as numerical-isolation fixtures, while their literature-point Option-A section uses the current production atomic dissipator and shared Maxwell/Q constructors. Its Floquet assembly remains analysis-owned as a cross-check, while production independently supplies general-order continued-fraction core and compiled-kernel paths. It also reconstructs M_E, Q, and T_E independently from literal SI prefactors and `scipy.linalg.expm`. At $N_F = 3$ the production/reference maximum differences are 1.64×10^{-14} or smaller. Tables 12, 18, and 19 are generated from that checked path. Passive Beer–Lambert, unrotated-to-symmetric mismatch equivalence, zero-coupling identity, constant-segment parity, the structural normalization ledger, an independent stored-χ Option-A transfer reference, Bogoliubov Q conversion, and Manley–Rowe energy bounds are checked in `tests/test_fwm_propagation.py`. General-order continued-fraction versus dense-block parity, all-harmonic trace and residual invariants, compiled-kernel parity for $N_F = 1, 2, 3$, and the full-scan convergence gate are checked in `tests/test_fwm_floquet.py`. The pump-only state diagnostics, static-frame/Floquet parity, finite-seed limit, projected-DC response, and direct pole/residue parity are emitted by `convergence_audit.py`; their solver-level fixtures are checked independently in `tests/test_fwm_pump_response.py`. The lab-frequency contract, collinear and equal-k limits, analytic rms, and two-dimensional grid/cutoff checks are recorded in `tests/test_fwm_angular_doppler.py`. The detailed analytic-identity table, full pole-spectrum table, and physicality-margin table remain report derivations or separately reported checks and are not described as regenerated by the audit script. Table 7 and the conclusion state the validation status of each model and calculation.

I Numerical reference data for the microscopic constructions

The equations, assumptions, model inputs and representative results are given in this document. The following machine-readable data provide full matrices, frequency grids and numerical residuals for the stated examples. They are numerical references within the declared models, not independent experimental measurements. Paths are relative to the repository root; version suffixes in data filenames identify immutable numerical artifacts.

- Atomic ordered noise, field elimination and normalization:
  `docs/grand_challenge/s1_atomic_noise_report.json`,
  `docs/grand_challenge/s1_field_noise_report.json`, and
  `docs/grand_challenge/s1_normalization_report.json`.
- Finite seeds, periodic QRT, readout and spatial modes:
  `docs/grand_challenge/s1_seed_validity_report_v2.json`,
  `docs/grand_challenge/s1_periodic_noise_report_v2.json`,
  `docs/grand_challenge/s1_periodic_field_report_v2.json`,
  `docs/grand_challenge/s1_readout_report_v2.json`, and
  `docs/grand_challenge/s1_spatial_field_report_v2.json`.
- Ballistic inflow, independent adjoint reference and step integration:
  `docs/grand_challenge/inflow_report_v1.json`,
  `docs/grand_challenge/adjoint_transport_report_v1.json`, and
  `docs/grand_challenge/exponential_transport_report_v1.json`.
- Thermal-grid and scramble data: `docs/grand_challenge/thermal_campaign_v2/`, with the stated $(p, \text{seed}) = (2, 11), (2, 211), (2, 811)$ conditions. Full matrix comparisons, rather than rounded scalar extrema, define the gate in Table 5.
- Independent random-generator QRT and boundary-sensitivity probe:
  `analysis/grand_challenge/checkpoint_2026_09_18/physics_probe.json` and its adjacent plain-text reproducer. The random seed is 20260918; Python 3.14.2, NumPy 2.4.1 and SciPy 1.17.0 were used. Inputs and source SHA-256 hashes are embedded in the data.

J Explicit countermodels: algebraic closure without physical validity

Numerical agreement between a transfer matrix and its Gaussian completion cannot repair an inconsistent optical model. To make this distinction quantitative, consider the diagnostic variant that retains a duplicate manifold population factor and uses the opposite symmetric-frame mismatch sign. Its static drift and transfer are

$$M_D = \begin{pmatrix} -1.330328 + 242.349727\mathrm{i} & -0.410000 + 195.707754\mathrm{i} \\ 0.693784 - 195.680004\mathrm{i} & 0.341352 - 171.013770\mathrm{i} \end{pmatrix} \mathrm{m}^{-1}, \tag{130}$$

$$T_D = \begin{pmatrix} -0.376876 + 2.339296\mathrm{i} & -0.929014 + 1.950994\mathrm{i} \\ 0.931714 - 1.949376\mathrm{i} & 1.593377 - 1.777683\mathrm{i} \end{pmatrix}. \tag{131}$$

Here equal photon energies and equal mode areas are approximated, so Q is proportional to the identity. The two squared transfer coefficients are 5.61434 and 4.66816, with gap 0.946182. Its commutator-defect matrix K has eigenvalues 0.189391 and 3.153969. Omitting a reservoir gives a global commutator residual 0.0551; the independent-vacuum completion reduces the local and global residuals to 4.44×10^{-16} and 1.04×10^{-15}. These small commutator residuals do not validate the duplicate population or mismatch sign.

Table 26: Noise diagnostics for the deliberately inconsistent static variant. External efficiency is 0.8694. Values are dimensionless ratios; numbers in parentheses are dB. None is an apparatus prediction.

Model and subtraction	Source ratio	Detected ratio
Chosen independent-vacuum completion, $w = 1$	0.09230 (−10.35)	0.21084 (−6.76)
Same completion, DC balance $w = 1.20269$	0.09774 (−10.10)	0.21558 (−6.67)
Ideal canonical amplifier matched to $G_p = 5.61434$	0.09776 (−10.10)	0.21560 (−6.66)

The ideal comparison has conjugate gain 4.61434 and therefore is a different channel. The static completion gives the same values at $0.1, 0.15, 2, 4\,\mathrm{MHz}$, so it cannot explain a spectral bandwidth or an electronic-noise peak. Local logarithmic sensitivities of this variant are $\partial \ln G_p / \partial \ln \ell_s \simeq 3.53$ and $\partial \ln G_c / \partial \ln \ell_s \simeq 4.29$. Changing its coherence-only ground damping from 90 to 110 kHz moves G_p from 5.6138 to 5.6033 and G_c from 4.6631 to 4.6573. These derivatives are conditional on the inconsistent variant and do not determine uncertainty in the reciprocal model. Against approximate comparison gains 15.5 and 13.6, its discrepancies would be −63.8% and −65.7%; the exact raw-power experimental comparison uses the separately specified seed powers in Table 20.

The gain-only covariance ansatz

$$C_{pc}^{\mathrm{ansatz}} = \tfrac{1}{2}\left[(G_p + g_c^{(\Phi)}) - (G_p - g_c^{(\Phi)})^2\right] \tag{132}$$

forces the scalar score in Eq. (121); it does not derive an ordered atomic spectrum or ensure uncertainty/CP constraints. Negative diagnostic values such as −8.10 or −15.62 dB computed from gain alone remain scores under that ansatz. The passive example $(G_p, g_c^{(\Phi)}) = (0.2, 0.1)$ already falsifies its use as a universal squeezing formula. Clipping the score into $[0, 1]$ cannot supply missing vacuum noise.

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